

Quantum-to-classical Lax correspondence for Class 5 and Class 6 models

1 Introduction

The Landau–Lifshitz model relates the long-wavelength dynamics of quantum spin chains to classical field theory. It also appears in limits of string sigma models: the coherent-state continuum action of the Heisenberg chain agrees with an action obtained for strings carrying large angular momentum [1]. Integrable deformations extend this relation to anisotropic Landau–Lifshitz models. Their classical Lax pairs and conserved-charge algebras have been studied in limits of deformed string backgrounds [2]. These connections motivate an explicit account of how the Hamiltonian and Lax structures pass from the quantum chain to the continuum theory.

The Class 5 and Class 6 models provide two non-Hermitian deformations of the spin- $\frac{1}{2}$ Heisenberg XXX chain. They arose in the classification of nearest-neighbour integrable chains in Ref. [3] and have transfer matrices with nontrivial Jordan blocks. De Leeuw, Fontanella and Nieto García derived their deformed Landau–Lifshitz continuum models and constructed higher conserved charges using a boost functional [4]. For Class 5, they also identified the quantum deformation as a Jordanian Drinfeld twist. The coexistence of this algebraic structure with a non-Hermitian Hamiltonian makes these chains useful examples for studying the relation between quantum and classical integrability.

Several classical constructions are available for this comparison. Avan, Doikou and Sfetsos formulated continuum limits that retain the classical Lax and r -matrix structures [5]. The relation between continuum Hamiltonians, time Lax matrices and boundary conditions was developed further in Ref. [6]. For Class 6, the quantum model is identified with the eleven-vertex model in Ref. [7], and its Landau–Lifshitz Lax formulation is already contained in the eleven-vertex construction of Levin, Olshanetsky and Zotov [8].

The question addressed here is how the quantum lattice *time* Lax operator produces the continuum time matrix for the Hamiltonian flow. In Section 2 of Ref. [4], the authors discuss a spatial Lax matrix for Class 5 and leave the construction of a compatible time component open. Taking the time operator directly to the continuum identifies the quantum terms that enter this component and permits a comparison with the classical r -matrix construction.

We perform this limit at fixed spin $\frac{1}{2}$, using the coherent-state lower symbol: the matrix element of an operator in a product of spin coherent states, with auxiliary-space indices retained. The lattice zero-curvature equation contains operator products whose factors share a physical site. At that site, the lower symbol of a product generally differs from the product of the separate symbols, as in standard coherent-state symbol calculus [9, 10]. Smooth variation of the spin labels between neighbouring sites controls the long-wavelength expansion, while products on a common site retain the fixed spin- $\frac{1}{2}$ operator algebra. Their contribution must therefore be evaluated at the order of the continuum time evolution.

For both chains, we compute this surviving contribution and absorb it into a local correction of the direct time symbol. In Class 5 it is a total spatial derivative; Class 6 also has a commutator contribution involving the second spatial-operator coefficient. We derive

the Hamiltonian equations independently and establish their equivalence to the corrected zero-curvature equations by identities that hold before imposing the equations of motion. Explicit inverse relations recover every component of the equations from the curvature. The undeformed XXX limit and the independent classical r -matrix construction for Class 5 provide further checks.

We also determine Lax pairs through numerical coefficient searches, following the optimization approach of Krippendorf, Lüst and Syvaeri [11]. The input consists of sampled local spin values and derivatives, and the coefficients of finite ansätze are fitted by minimizing off-shell curvature residuals. The tensor structures are specified using the null direction in Class 5 and the nilpotent anisotropy in Class 6. In Class 5, the coefficients of eight additional candidate terms were suppressed; in Class 6, the recovered relations revealed a finite matrix-square-root expression. Appendix A gives the ansätze, loss functions, numerical settings and analytic verification.

Section 2 fixes the lattice conventions and formulates the continuum problem. Sections 3 and 4 carry out the two model calculations. Section 5 compares the surviving corrections and their dependence on the lattice operators. We summarize the results and discuss their scope and possible extensions in Section 6.

2 Quantum lattice setup and continuum limit

This section formulates the continuum problem common to the two spin chains. We first fix the quantum lattice Lax equation and then define the coherent-state lower symbols and long-wavelength scaling. Keeping the contributions from products on overlapping physical sites leads to the condition that determines the correction of the continuum time Lax matrix.

2.1 Quantum spin chains and the lattice Lax equation

The two lattice models studied in this paper are homogeneous periodic spin chains with nearest-neighbour interactions and a two-dimensional local Hilbert space. We use the standard quantum inverse-scattering framework for such chains [12, 5], with the following notation.

Let $\mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2$ denote a reference local Hilbert space. Consider a chain with N lattice sites, and let $\mathcal{H}_n$ be a copy of $\mathcal{H}_{\text{loc}}$ assigned to the n th site,

$$\mathcal{H}_n \simeq \mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2, \quad n = 1, \dots, N. \quad (2.1)$$

The Hilbert space of the full chain is the tensor product

$$\mathcal{H}^{(N)} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_N. \quad (2.2)$$

We impose periodic boundary conditions, so the site labels are understood modulo N ; in particular, the site following N is site 1.

A homogeneous nearest-neighbour Hamiltonian is specified by a single *two-site Hamiltonian operator*

$$h \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (2.3)$$

The adjective “homogeneous” means that the same two-site operator h is used on every nearest-neighbour bond. We denote by $h_{n,n+1}$ the operator on the full Hilbert space (2.2) that acts as h on $\mathcal{H}_n \otimes \mathcal{H}_{n+1}$ and as the identity on all remaining sites. For $1 \leq n < N$, this embedding is explicitly

$$h_{n,n+1} = \mathbf{1}_{\mathcal{H}_1} \otimes \dots \otimes \mathbf{1}_{\mathcal{H}_{n-1}} \otimes h \otimes \mathbf{1}_{\mathcal{H}_{n+2}} \otimes \dots \otimes \mathbf{1}_{\mathcal{H}_N}. \quad (2.4)$$

For the bond joining the last and first sites, $h_{N,1}$ denotes the same operator h acting on $\mathcal{H}_N \otimes \mathcal{H}_1$, with the other sites left unchanged. Thus $h_{n,n+1}$ is the copy of the fixed two-site operator h on the n th bond.

With these conventions, the Hamiltonian of the periodic chain is

$$H^{\text{lat}} = \sum_{n=1}^N h_{n,n+1}, \quad h_{N,N+1} \equiv h_{N,1}. \quad (2.5)$$

The model dependence therefore enters through the choice of the single matrix h . We next introduce the quantum R -matrix and the lattice Lax equation associated with a nearest-neighbour integrable chain.

For the integrable chains considered in this paper, the local interaction is encoded by a quantum R -matrix. Introduce an auxiliary space

$$\mathcal{V}_a \simeq \mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2, \quad (2.6)$$

which is a tensor factor distinct from the physical Hilbert spaces $\mathcal{H}_n$. When several auxiliary spaces are required, we label them by a, b, c .

Let $u \in \mathbb{C}$ denote the *quantum spectral parameter*. A quantum R -matrix in the representation used here is a family of two-site operators

$$R(u) \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (2.7)$$

Whenever two tensor factors are individually labelled, the corresponding subscripts indicate the copy of this same two-site operator acting on those factors. Thus, for example, $R_{ab}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{V}_b)$ and $R_{a,n}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{H}_n)$.

The models considered here have an R -matrix of difference form. Its Yang–Baxter equation is

$$\begin{aligned} R_{ab}(u-v) R_{ac}(u-w) R_{bc}(v-w) \\ = R_{bc}(v-w) R_{ac}(u-w) R_{ab}(u-v), \end{aligned} \quad (2.8)$$

which is an operator identity on $\mathcal{V}_a \otimes \mathcal{V}_b \otimes \mathcal{V}_c$ [12]. The subscripts indicate the two tensor factors on which each copy of R acts.

Let

$$P \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}) \quad (2.9)$$

be the permutation operator defined by $P(|v\rangle \otimes |w\rangle) = |w\rangle \otimes |v\rangle$. As for the R -matrix, P_{ab} denotes its copy on $\mathcal{V}_a \otimes \mathcal{V}_b$. We normalize the R -matrix so that its regular point is $u = 0$,

$$R(0) = P, \quad \text{equivalently} \quad R_{ab}(0) = P_{ab}. \quad (2.10)$$

For a regular fundamental spin- $\frac{1}{2}$ chain, the two-site Hamiltonian operator introduced in (2.3) is obtained from the same two-site R -matrix as

$$h = P \left. \frac{\partial R(u)}{\partial u} \right|_{u=0}. \quad (2.11)$$

This is the standard relation between a regular R -matrix and the nearest-neighbour Hamiltonian density in the quantum inverse-scattering construction [12, 3].

A scalar rescaling $R(u) \mapsto f(u)R(u)$ with $f(0) = 1$ preserves the regularity convention (2.10) and adds $f'(0)\mathbf{1}$ to h . This term shifts the total energy by a constant and leaves the Heisenberg equations unchanged. An overall rescaling of the spectral parameter amounts to

a corresponding normalization of the Hamiltonian. These normalization freedoms are fixed together with each explicit R -matrix.

To associate the R -matrix with a particular physical site, one copy of its second tensor factor is identified with the local Hilbert space $\mathcal{H}_n$. The resulting *quantum lattice Lax operator* is

$$L_{a,n}(u) := R_{a,n}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{H}_n). \quad (2.12)$$

For a fixed physical site n , the Yang–Baxter equation implies the local exchange relation

$$R_{ab}(u-v)L_{a,n}(u)L_{b,n}(v) = L_{b,n}(v)L_{a,n}(u)R_{ab}(u-v), \quad (2.13)$$

acting on $\mathcal{V}_a \otimes \mathcal{V}_b \otimes \mathcal{H}_n$.

The ordered product of lattice Lax operators around the periodic chain defines the *quantum monodromy matrix*

$$T_a(u) := L_{a,N}(u)L_{a,N-1}(u) \cdots L_{a,1}(u), \quad (2.14)$$

and tracing over the auxiliary space gives the *transfer matrix*

$$\tau(u) := \text{tr}_a T_a(u). \quad (2.15)$$

Equation (2.13) implies

$$[\tau(u), \tau(v)] = 0, \quad (2.16)$$

so an expansion of $\tau(u)$ generates mutually commuting quantum charges [12]. At the regular point, the logarithmic derivative of the transfer matrix reproduces the nearest-neighbour Hamiltonian (2.5), up to the scalar and normalization freedoms described after (2.11).

We next need the local equation governing the time evolution of $L_{a,n}(u)$. Differentiating the Yang–Baxter relation at its regular point and using (2.11) gives the *Sutherland relation* [3]

$$\begin{aligned} [h_{n-1,n}, L_{a,n}(u)L_{a,n-1}(u)] \\ = L_{a,n}(u) \partial_u L_{a,n-1}(u) - (\partial_u L_{a,n}(u)) L_{a,n-1}(u). \end{aligned} \quad (2.17)$$

Here $h_{n-1,n}$ denotes the embedding, defined above, of the same two-site Hamiltonian h on the neighbouring physical sites $n-1$ and n .

Let t_{lat} denote the quantum lattice time. For an operator $\mathcal{O}$ with no explicit time dependence we use the Heisenberg equation

$$\partial_{t_{\text{lat}}} \mathcal{O} = i[H^{\text{lat}}, \mathcal{O}]. \quad (2.18)$$

Because $L_{a,n}$ acts only on the physical site n , only the two Hamiltonian terms $h_{n-1,n}$ and $h_{n,n+1}$ contribute to its commutator with H^{lat} . The Sutherland relation organizes these two contributions into a discrete zero-curvature equation.

For spectral parameters at which $L_{a,n}(u)$ is invertible, define the *finite-lattice time Lax operator*

$$\begin{aligned} A_{a,n}(u) = i h_{n-1,n} - i L_{a,n}(u)^{-1} h_{n-1,n} L_{a,n}(u) \\ - i L_{a,n}(u)^{-1} \partial_u L_{a,n}(u). \end{aligned} \quad (2.19)$$

The physical part of $A_{a,n}$ acts on the neighbouring sites $n-1$ and n , while its matrix indices belong to the auxiliary space $\mathcal{V}_a$. Equations (2.17) and (2.18) then give

$$\partial_{t_{\text{lat}}} L_{a,n}(u) = A_{a,n+1}(u) L_{a,n}(u) - L_{a,n}(u) A_{a,n}(u). \quad (2.20)$$

This is the lattice analogue of a zero-curvature equation. The time Lax operator has a scalar ambiguity: adding to every $A_{a,n}$ the same scalar function of u times the identity on the full tensor product leaves (2.20) unchanged. This freedom permits convenient normalizations.

The continuum limit of (2.20) depends on the physical support of the operators in its two products. They contain operators that share a physical lattice site: $A_{a,n+1}$ and $L_{a,n}$ both act on $\mathcal{H}_n$, while $L_{a,n}$ and $A_{a,n}$ also overlap on $\mathcal{H}_n$. Consequently, coherent-state expectation values of these products retain correlations associated with the shared physical site.

2.2 Coherent states and long-wavelength limit

We next describe the common kinematical ingredients used to pass from the spin chain to a smooth classical spin field. We use standard $SU(2)$ spin coherent states [9]; their use in long-wavelength limits of integrable spin chains is reviewed, for example, in [5]. The same construction is used in the direct continuum analysis of the Class 5 and Class 6 chains [4].

Let $\sigma^1, \sigma^2, \sigma^3$ denote the Pauli matrices on the reference local Hilbert space $\mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2$, and let $|\uparrow\rangle$ and $|\downarrow\rangle$ be the eigenvectors of σ^3 with eigenvalues $+1$ and -1 , respectively. The spin- $\frac{1}{2}$ operators are normalized as

$$s^i := \frac{1}{2}\sigma^i, \quad i = 1, 2, 3. \quad (2.21)$$

Throughout, repeated Cartesian indices $i, j, k \in \{1, 2, 3\}$ are summed. At the n th lattice site, a normalized spin coherent state is parametrized by polar and azimuthal angles θ_n and ϕ_n as

$$|\Omega_n\rangle = \cos \frac{\theta_n}{2} |\uparrow\rangle + e^{i\phi_n} \sin \frac{\theta_n}{2} |\downarrow\rangle. \quad (2.22)$$

This phase convention fixes the associated unit vector. We define the *classical spin vector at site n* by

$$\mathbf{S}_n := \langle \Omega_n | \boldsymbol{\sigma} | \Omega_n \rangle = \begin{pmatrix} \sin \theta_n \cos \phi_n \\ \sin \theta_n \sin \phi_n \\ \cos \theta_n \end{pmatrix}, \quad \boldsymbol{\sigma} := (\sigma^1, \sigma^2, \sigma^3)^\top. \quad (2.23)$$

It follows directly that

$$\mathbf{S}_n^\top \mathbf{S}_n = 1, \quad \langle \Omega_n | s^i | \Omega_n \rangle = \frac{1}{2} S_n^i. \quad (2.24)$$

Thus the normalization of the classical spin field is fixed independently of the model-dependent normalization of the Lax matrix.

For a configuration of the full chain we use the normalized product coherent state

$$|\Omega\rangle := |\Omega_1\rangle \otimes \cdots \otimes |\Omega_N\rangle \in \mathcal{H}^{(N)}. \quad (2.25)$$

The coherent-state labels are allowed to depend on time. We denote by t the time coordinate of the continuum field theory. Its relation to the quantum lattice time t_{lat} introduced in (2.18) is part of the continuum scaling.

Let $\epsilon > 0$ denote the *lattice spacing*. We assign the spatial coordinate

$$x_n := (n-1)\epsilon, \quad n = 1, \dots, N, \quad (2.26)$$

and take the long-wavelength limit

$$\epsilon \rightarrow 0, \quad N \rightarrow \infty, \quad \ell := N\epsilon \text{ fixed}, \quad (2.27)$$

where ℓ is the physical circumference of the continuum spatial circle. The long-wavelength assumption is that the coherent-state labels vary smoothly from one lattice site to the next. Equivalently, there is a smooth unit spin field $\mathbf{S}(x, t)$ such that

$$\mathbf{S}_n(t) = \mathbf{S}(x_n, t), \quad \mathbf{S}(x + \ell, t) = \mathbf{S}(x, t). \quad (2.28)$$

For neighbouring sites this gives the Taylor expansion

$$\mathbf{S}_{n+1} = \mathbf{S} + \epsilon \partial_x \mathbf{S} + \frac{\epsilon^2}{2} \partial_x^2 \mathbf{S} + \mathcal{O}(\epsilon^3), \quad (2.29)$$

where the fields on the right-hand side are evaluated at (x_n, t) . Differentiating the unit-spin constraint $\mathbf{S}^\top \mathbf{S} = 1$ gives the kinematical identities

$$\mathbf{S}^\top \partial_x \mathbf{S} = 0, \quad \mathbf{S}^\top \partial_x^2 \mathbf{S} = -(\partial_x \mathbf{S})^\top (\partial_x \mathbf{S}). \quad (2.30)$$

For any smooth scalar density $f(x)$, the corresponding lattice sum obeys the usual Riemann-sum relation

$$\epsilon \sum_{n=1}^N f(x_n) \longrightarrow \int_0^\ell dx f(x). \quad (2.31)$$

The powers of ϵ that accompany the Hamiltonian, deformation parameters, and time coordinate are model dependent.

We now define the operation that maps local quantum operators to classical quantities. Let X_a be an operator acting on the auxiliary space $\mathcal{V}_a$ and on any finite set of physical lattice sites. Its *coherent-state lower symbol*, denoted by the superscript $\downarrow$ [13], is the auxiliary-space operator

$$X_a^\downarrow := (\mathbf{1}_{\mathcal{V}_a} \otimes \langle \Omega |) X_a (\mathbf{1}_{\mathcal{V}_a} \otimes | \Omega \rangle). \quad (2.32)$$

Only the physical sites on which X_a acts contribute to the matrix element; the remaining factors in (2.25) cancel by normalization. If X has no auxiliary-space indices, $X^\downarrow$ is a scalar function of the coherent-state labels. In particular, the lower symbol of the lattice Lax operator is

$$L_{a,n}^\downarrow(u) = \langle \Omega_n | L_{a,n}(u) | \Omega_n \rangle, \quad (2.33)$$

which is a 2×2 matrix acting on the auxiliary space.

After evaluating these symbols on real spin vectors, we extend their algebraic expressions to complex spin components satisfying $\mathbf{S}^\top \mathbf{S} = 1$. Here $\top$ denotes the transpose. Scalar and vector products are bilinear on this complexified spin space.

The product of two Pauli matrices on the same site satisfies $\sigma^i \sigma^j = \delta_{ij} \mathbf{1}_2 + i \varepsilon_{ijk} \sigma^k$, where δ_{ij} is the Kronecker delta and ε_{ijk} is completely antisymmetric with $\varepsilon_{123} = 1$. Thus

$$(\sigma^i \sigma^j)^\downarrow - (\sigma^i)^\downarrow (\sigma^j)^\downarrow = \delta_{ij} - S^i S^j + i \varepsilon_{ijk} S^k. \quad (2.34)$$

Together with factorization between distinct physical sites, this identity evaluates all overlapping operator products used below.

The continuum spatial Lax matrix is extracted after the model-dependent normalization and spectral scaling have been fixed. Let λ denote the *continuum Lax spectral parameter*, to distinguish it from the quantum spectral parameter u defined in Section 2.1. For each model we choose a nonzero scalar normalization $c_L(\epsilon, \lambda)$ and a relation $u = u(\epsilon, \lambda)$ such that the normalized lattice operator

$$\widehat{L}_{a,n}(\epsilon, \lambda) := c_L(\epsilon, \lambda) L_{a,n}(u(\epsilon, \lambda)) \quad (2.35)$$

has a regular long-wavelength expansion. When its lower symbol takes the form

$$\widehat{L}_{a,n}^\downarrow(\epsilon, \lambda) = \mathbf{1}_{\mathcal{V}_a} + \epsilon U(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad (2.36)$$

we call the coefficient $U(x, t; \lambda)$ the *continuum spatial Lax matrix*. The choices of c_L , of the scaling $u(\epsilon, \lambda)$, and of any deformation parameters are part of the model-specific continuum limit; they are specified separately for the two models.

The discrete zero-curvature equation (2.20) also contains products of operators with overlapping physical support. We next evaluate their lower symbols and determine their contribution to the continuum zero-curvature equation.

2.3 From the lattice Lax equation to the continuum zero-curvature equation

We now apply the lower-symbol operation to the finite-lattice equation (2.20). At this stage the lower symbol is taken at a fixed point of the coherent-state phase space. Applying it to the operator identity gives the exact relation

$$\begin{aligned} (\partial_{t_{\text{lat}}} L_{a,n}(u))^\downarrow &= (A_{a,n+1}(u) L_{a,n}(u))^\downarrow \\ &\quad - (L_{a,n}(u) A_{a,n}(u))^\downarrow. \end{aligned} \quad (2.37)$$

For each model, we derive the continuum Hamiltonian flow independently and use it to identify the continuum limit of the left-hand side with the time derivative of the classical spatial Lax matrix. The factors in each product on the right-hand side act on a common physical site. The lower symbol of each product contains a contribution in addition to the product of the separate lower symbols. We denote these contributions by the auxiliary-space matrices

$$\mathcal{D}_{a,n}^{(+)}(u) := (A_{a,n+1}(u) L_{a,n}(u))^\downarrow - A_{a,n+1}^\downarrow(u) L_{a,n}^\downarrow(u), \quad (2.38)$$

$$\mathcal{D}_{a,n}^{(-)}(u) := (L_{a,n}(u) A_{a,n}(u))^\downarrow - L_{a,n}^\downarrow(u) A_{a,n}^\downarrow(u). \quad (2.39)$$

These matrices measure the failure of the lower symbol to preserve the operator product at the overlapping physical site. This nonmultiplicativity is standard in coherent-state symbol calculus and is commonly encoded by a Berezin star product [10].

With the definitions (2.38) and (2.39), the lower-symbol form of the lattice Lax equation is the exact identity

$$\begin{aligned} (\partial_{t_{\text{lat}}} L_{a,n}(u))^\downarrow &= A_{a,n+1}^\downarrow(u) L_{a,n}^\downarrow(u) - L_{a,n}^\downarrow(u) A_{a,n}^\downarrow(u) \\ &\quad + \mathcal{D}_{a,n}^{(+)}(u) - \mathcal{D}_{a,n}^{(-)}(u). \end{aligned} \quad (2.40)$$

For operators with disjoint physical support, the product coherent state (2.25) makes the corresponding difference vanish. The difference terms in (2.40) can survive because of the overlap described at the end of Section 2.1.

We next state the common continuum scaling used by the two models studied in this paper. The model-dependent spectral scaling and scalar normalization of the spatial Lax operator have already been represented abstractly by (2.35). We also use the freedom described below (2.20) to shift the finite-lattice time Lax operator by a scalar function times the identity. We denote the resulting operator by $\widehat{A}_{a,n}$.

For both models, the continuum time coordinate t is related to the quantum lattice time t_{lat} by

$$t = \epsilon^2 t_{\text{lat}}. \quad (2.41)$$

Equation (2.41) fixes the normalization of the continuum time used in this paper. Since the scalar normalization of $\hat{L}_{a,n}$ and the scalar shift defining $\hat{A}_{a,n}$ are independent of t_{lat} and of the lattice site, the normalized operators obey the same discrete zero-curvature equation as $L_{a,n}$ and $A_{a,n}$. After the model-dependent spectral scaling has been imposed, the lower symbols of the normalized spatial and time Lax operators have the form

$$\hat{L}_{a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad (2.42)$$

$$\hat{A}_{a,n}^\downarrow = \epsilon^2 V^\downarrow(x_n, t; \lambda) + \mathcal{O}(\epsilon^3). \quad (2.43)$$

Here $V^\downarrow(x, t; \lambda)$ denotes the continuum matrix obtained directly from the lower symbol of the finite-lattice time Lax operator. The scalings (2.41)–(2.43) are checked for each explicit model.

The part of (2.40) containing only products of lower symbols now has a standard continuum limit. Using $x_{n+1} = x_n + \epsilon$ together with (2.42) and (2.43), one finds

$$\begin{aligned} \hat{A}_{a,n+1}^\downarrow \hat{L}_{a,n}^\downarrow - \hat{L}_{a,n}^\downarrow \hat{A}_{a,n}^\downarrow \\ = \epsilon^3 \left(\partial_x V^\downarrow + [V^\downarrow, U] \right)_{(x_n, t; \lambda)} + \mathcal{O}(\epsilon^4). \end{aligned} \quad (2.44)$$

The part containing products of lower symbols starts at order ϵ^3 . The coefficient of the lower symbol of the Heisenberg derivative at the same order is determined from the continuum Hamiltonian and its equations of motion for each model. This dynamical step is separate from the algebraic treatment of the operator products.

For the normalized operators, define the corresponding differences by

$$\hat{\mathcal{D}}_{a,n}^{(+)} := (\hat{A}_{a,n+1} \hat{L}_{a,n})^\downarrow - \hat{A}_{a,n+1}^\downarrow \hat{L}_{a,n}^\downarrow, \quad (2.45)$$

$$\hat{\mathcal{D}}_{a,n}^{(-)} := (\hat{L}_{a,n} \hat{A}_{a,n})^\downarrow - \hat{L}_{a,n}^\downarrow \hat{A}_{a,n}^\downarrow. \quad (2.46)$$

Their continuum contribution, when the limit exists, is

$$\mathcal{C}(x, t; \lambda) := \lim_{\epsilon \rightarrow 0} \frac{\hat{\mathcal{D}}_{a,n}^{(+)} - \hat{\mathcal{D}}_{a,n}^{(-)}}{\epsilon^3}, \quad x = x_n. \quad (2.47)$$

The evaluation of this limit requires both the operator expansion and the Taylor expansion of the spin arguments. Let $m \in \{5, 6\}$ label the model, and write $\mathcal{C}_m$ for its continuum contribution. For independent spin vectors $\mathbf{S}_L$ and $\mathbf{S}_R$ on the left and right physical sites, define the auxiliary-space matrix functions $\mathcal{D}_{m,2}^{(\pm)}$ and $\mathcal{D}_{m,3}^{(\pm)}$ as the coefficients

$$\hat{\mathcal{D}}_{m;a,n}^{(+)} = \epsilon^2 \mathcal{D}_{m,2}^{(+)}(\mathbf{S}_n, \mathbf{S}_{n+1}) + \epsilon^3 \mathcal{D}_{m,3}^{(+)}(\mathbf{S}_n, \mathbf{S}_{n+1}) + \mathcal{O}(\epsilon^4), \quad (2.48)$$

$$\hat{\mathcal{D}}_{m;a,n}^{(-)} = \epsilon^2 \mathcal{D}_{m,2}^{(-)}(\mathbf{S}_{n-1}, \mathbf{S}_n) + \epsilon^3 \mathcal{D}_{m,3}^{(-)}(\mathbf{S}_{n-1}, \mathbf{S}_n) + \mathcal{O}(\epsilon^4). \quad (2.49)$$

The dependence on the spectral parameter and continuum coupling is suppressed. The final subscript denotes the power of ϵ in the operator-product difference before expanding the spin arguments. For both models, the quadratic coefficients agree at coincident spins, $\mathcal{D}_{m,2}^{(+)}(\mathbf{S}, \mathbf{S}) = \mathcal{D}_{m,2}^{(-)}(\mathbf{S}, \mathbf{S})$, as verified in the model calculations below. Taylor expansion at $x = x_n$ then gives

$$\begin{aligned} \mathcal{C}_m = \partial_x S^i \left(\frac{\partial \mathcal{D}_{m,2}^{(+)}}{\partial S_R^i} + \frac{\partial \mathcal{D}_{m,2}^{(-)}}{\partial S_L^i} \right) \Big|_{\mathbf{S}_L = \mathbf{S}_R = \mathbf{S}} \\ + \mathcal{D}_{m,3}^{(+)}(\mathbf{S}, \mathbf{S}) - \mathcal{D}_{m,3}^{(-)}(\mathbf{S}, \mathbf{S}). \end{aligned} \quad (2.50)$$

The Cartesian derivatives act on the polynomial lower symbols before imposing the unit-spin constraints and before setting the two arguments equal. Both derivatives have a plus sign: the shift $\mathbf{S}_{n-1} = \mathbf{S} - \epsilon \partial_x \mathbf{S} + \dots$ occurs in the term subtracted in Eq. (2.47). Equation (2.50) separates the contribution from the spatial variation of the quadratic coefficients from the cubic coefficients at coincident spins.

Once the model-specific Hamiltonian check identifies the continuum limit of the lower symbol of the Heisenberg derivative on the left-hand side of (2.40) with $\partial_t U$, the order- ϵ^3 equation becomes

$$\partial_t U = \partial_x V^\downarrow + [V^\downarrow, U] + \mathcal{C}. \quad (2.51)$$

The continuum zero-curvature equation is recovered if the matrix $\mathcal{C}$ can be written in the local form

$$\mathcal{C} = \partial_x \Delta V + [\Delta V, U] \quad (2.52)$$

for some auxiliary-space matrix $\Delta V(x, t; \lambda)$. In that case the continuum time Lax matrix

$$V := V^\downarrow + \Delta V \quad (2.53)$$

satisfies

$$\partial_t U - \partial_x V + [U, V] = 0. \quad (2.54)$$

Equations (2.47)–(2.54) formulate the continuum problem. The matrix ΔV is determined from the finite-lattice operator products for each model.

3 Class 5 model

We apply the construction of Section 2 to the Class 5 chain. After fixing the quantum model and spatial continuum limit, we evaluate the finite-lattice time operator and its overlapping products. Their contribution to the continuum Lax equation is a total spatial derivative. We then compare the corrected Lax equation with the Hamiltonian equations of motion and construct the time matrix independently from the classical r -matrix.

3.1 Quantum Class 5 model and spatial Lax matrix

The name “Class 5” refers to one of the regular 4×4 difference-form solutions of the Yang–Baxter equation found in the classification of nearest-neighbour integrable spin- $\frac{1}{2}$ chains in Ref. [3]. The long-wavelength limit of this chain is the deformed Landau–Lifshitz model studied in Ref. [4]. We use a one-parameter representative of the quantum chain for the quantum-to-classical Lax calculation. This representative is related to the general Class 5 R -matrix by the spectral normalization and local basis transformation described in Ref. [4].

Let $\kappa_5 \in \mathbb{C}$ denote the *Class 5 lattice deformation parameter*. It corresponds to the parameter denoted by a in Ref. [4]. In the ordered two-spin basis

$$\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}, \quad (3.1)$$

the two-site Hamiltonian operator defining the Class 5 chain in our normalization is

$$h_5(\kappa_5) = \begin{pmatrix} 2 & 2\kappa_5 & -2\kappa_5 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \quad (3.2)$$

Following the embedding convention of Section 2.1, the copy of h_5 on the bond joining sites n and $n + 1$ is denoted by $h_{5;n,n+1}(\kappa_5)$. The periodic quantum Hamiltonian is therefore

$$H_5^{\text{lat}}(\kappa_5) = \sum_{n=1}^N h_{5;n,n+1}(\kappa_5), \quad h_{5;N,N+1} \equiv h_{5;N,1}. \quad (3.3)$$

For $\kappa_5 \neq 0$ this Hamiltonian is non-Hermitian in the standard inner product. The undeformed value $\kappa_5 = 0$ reduces, up to the normalization fixed in Section 2.1, to the spin- $\frac{1}{2}$ XXX chain.

The same normalization is generated by the regular Class 5 quantum R -matrix

$$R_5(u; \kappa_5) = \begin{pmatrix} 1+2u & 2\kappa_5 u & -2\kappa_5 u & 0 \\ 0 & 2u & 1 & 0 \\ 0 & 1 & 2u & 0 \\ 0 & 0 & 0 & 1+2u \end{pmatrix}, \quad (3.4)$$

where u is the quantum spectral parameter introduced in Section 2.1. It obeys the Yang–Baxter equation and the regularity condition

$$R_5(0; \kappa_5) = P, \quad (3.5)$$

and a direct differentiation gives

$$h_5(\kappa_5) = P \left. \frac{\partial R_5(u; \kappa_5)}{\partial u} \right|_{u=0}, \quad (3.6)$$

in agreement with the general relation (2.11). Equations (3.2) and (3.4) fix the model and the normalizations used in this section.

We isolate the deformation in the two-site operator K_5 , defined by

$$K_5 := \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (3.7)$$

With the permutation operator P defined in (2.9), the Class 5 data become

$$R_5(u; \kappa_5) = 2u\mathbf{1}_4 + P + 2\kappa_5 u K_5, \quad h_5(\kappa_5) = 2P + 2\kappa_5 K_5. \quad (3.8)$$

The operator K_5 is nilpotent and satisfies

$$K_5^2 = 0, \quad PK_5 = K_5, \quad K_5P = -K_5. \quad (3.9)$$

These identities simplify the finite-lattice time Lax operator.

We now specify the long-wavelength scaling. The lattice deformation is sent to zero together with the lattice spacing ϵ , while the *Class 5 continuum deformation coupling* α_5 is held fixed:

$$\kappa_5 = \epsilon \alpha_5. \quad (3.10)$$

This is the scaling used in the direct continuum construction of Ref. [4]; it places the deformation and the ordinary Landau–Lifshitz terms at the same order in the continuum action. For the Lax operator we simultaneously relate the quantum spectral parameter u to the continuum Lax spectral parameter λ by

$$u = \frac{\lambda}{\epsilon}, \quad \lambda \neq 0, \quad (3.11)$$

and use the common time scaling $t = \epsilon^2 t_{\text{lat}}$ introduced in (2.41).

For the fundamental spin- $\frac{1}{2}$ representation the lattice Lax operator is $L_{5;a,n}(u) := R_{5;a,n}(u; \kappa_5)$. We normalize it by its leading scalar part,

$$\widehat{L}_{5;a,n}(\epsilon, \lambda) := \frac{1}{2u} L_{5;a,n}(u). \quad (3.12)$$

The quantum spectral derivatives are taken at fixed κ_5 , before applying the scalings (3.10) and (3.11). The normalized operator has the exact expansion

$$\widehat{L}_{5;a,n} = \mathbf{1}_{\mathcal{V}_a \otimes \mathcal{H}_n} + \epsilon L_{5;a,n}^{(1)}, \quad L_5^{(1)} = \frac{P}{2\lambda} + \alpha_5 K_5. \quad (3.13)$$

Here $L_{5;a,n}^{(1)}$ denotes the copy of the fixed two-site operator $L_5^{(1)}$ on $\mathcal{V}_a \otimes \mathcal{H}_n$.

To evaluate its lower symbol, define the classical transverse spin combinations

$$S^+(x, t) := S^1(x, t) + iS^2(x, t), \quad S^-(x, t) := S^1(x, t) - iS^2(x, t). \quad (3.14)$$

Throughout this section, $S^\pm$ denotes these components of the classical unit spin field defined in (2.28). At the point $x = x_n$, the lower symbols of $P_{a,n}$ and $K_{5;a,n}$ are

$$P_{a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad (3.15)$$

$$K_{5;a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} S^+ & -(1 + S^3) \\ 0 & 0 \end{pmatrix}. \quad (3.16)$$

Substituting these expressions into (3.13) gives

$$\widehat{L}_{5;a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U_5(x_n, t; \lambda), \quad (3.17)$$

where the *Class 5 continuum spatial Lax matrix* is

$$U_5(x, t; \lambda) = \frac{1}{4\lambda} \begin{pmatrix} 1 + S^3 + 2\alpha_5 \lambda S^+ & S^- - 2\alpha_5 \lambda (1 + S^3) \\ S^+ & 1 - S^3 \end{pmatrix}. \quad (3.18)$$

The finite-lattice limit retains the complete $\mathfrak{gl}(2)$ matrix in Eq. (3.18), including its scalar part.

3.2 Time Lax matrix from the finite-lattice construction

We now take the long-wavelength limit of the finite-lattice time Lax operator. The starting point is the operator $A_{a,n}(u)$ defined in Eq. (2.19). Adding the same scalar multiple of the identity to every time Lax operator leaves the lattice zero-curvature equation unchanged. We use the representative

$$\widehat{A}_{5;a,n} := A_{5;a,n} + \frac{i}{u} \mathbf{1}, \quad (3.19)$$

where the identity acts on the auxiliary space and on the two physical sites $n-1, n$ supporting $A_{5;a,n}$. In terms of the normalized spatial operator $\widehat{L}_{5;a,n}$ in Eq. (3.12),

$$\widehat{A}_{5;a,n} = -i\widehat{L}_{5;a,n}^{-1} \left([h_{5;n-1,n}, \widehat{L}_{5;a,n}] + \partial_u \widehat{L}_{5;a,n} \right). \quad (3.20)$$

The derivative with respect to u is taken at fixed lattice deformation parameter κ_5 . After imposing the continuum scalings (3.10) and (3.11),

$$\partial_u \widehat{L}_{5;a,n} = -\frac{P_{a,n}}{2u^2} = -\epsilon^2 \frac{P_{a,n}}{2\lambda^2}. \quad (3.21)$$

The Hamiltonian on the neighbouring physical sites has the expansion

$$h_{5;n-1,n} = h_{n-1,n}^{(0)} + \epsilon h_{n-1,n}^{(1)}, \quad h^{(0)} := 2P, \quad h^{(1)} := 2\alpha_5 K_5. \quad (3.22)$$

Together with the exact relation $\hat{L}_{5;a,n} = \mathbf{1} + \epsilon L_{5;a,n}^{(1)}$ in Eq. (3.13), Eq. (3.20) gives

$$\hat{A}_{5;a,n} = \epsilon A_{5;a,n}^{(1)} + \epsilon^2 A_{5;a,n}^{(2)} + \mathcal{O}(\epsilon^3), \quad (3.23)$$

with

$$A_{5;a,n}^{(1)} = -i[h_{n-1,n}^{(0)}, L_{5;a,n}^{(1)}], \quad (3.24)$$

$$\begin{aligned} A_{5;a,n}^{(2)} &= iL_{5;a,n}^{(1)}[h_{n-1,n}^{(0)}, L_{5;a,n}^{(1)}] \\ &\quad - i[h_{n-1,n}^{(1)}, L_{5;a,n}^{(1)}] + \frac{i}{2\lambda^2} P_{a,n}. \end{aligned} \quad (3.25)$$

Both coefficients act on the tensor product of the auxiliary space and the two neighbouring physical sites.

We first evaluate the lower symbol of Eq. (3.23). Let $\mathbf{S}_{n-1}$ and $\mathbf{S}_n$ be the coherent-state spin vectors on the sites $n-1$ and n . Direct contraction of Eq. (3.24) gives

$$(A_{5;a,n}^{(1)})^\downarrow = -2(\mathbf{S}_{n-1} \times \mathbf{S}_n)^i \frac{\partial U_5}{\partial S^i}. \quad (3.26)$$

The Cartesian derivatives act on Eq. (3.18), with $S^\pm = S^1 \pm iS^2$, before imposing the unit-spin constraint. They are independent of the spin because U_5 is affine in its components. Using

$$\mathbf{S}_{n-1} = \mathbf{S}(x_n, t) - \epsilon \partial_x \mathbf{S}(x_n, t) + \mathcal{O}(\epsilon^2), \quad (3.27)$$

we have

$$\mathbf{S}_{n-1} \times \mathbf{S}_n = \epsilon \mathbf{S} \times \partial_x \mathbf{S} + \mathcal{O}(\epsilon^2). \quad (3.28)$$

The term $A_{5;a,n}^{(2)}$ already carries the prefactor ϵ^2 in Eq. (3.23); its lower symbol is therefore needed only at coincident spin arguments at this order. Using the lower symbols in Eqs. (3.15) and (3.16), together with $\mathbf{S}^\top \mathbf{S} = 1$, gives

$$(A_{5;a,n}^{(2)})^\downarrow \Big|_{\mathbf{S}_{n-1}=\mathbf{S}_n=\mathbf{S}} = 4iU_5^2 - \frac{i}{2\lambda^2} \mathbf{1}_{\mathcal{V}_a}. \quad (3.29)$$

Equations (3.26)–(3.29) then yield

$$\hat{A}_{5;a,n}^\downarrow = \epsilon^2 V_5^\downarrow(x_n, t; \lambda) + \mathcal{O}(\epsilon^3), \quad (3.30)$$

where

$$V_5^\downarrow = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_5}{\partial S^i} + 4iU_5^2 - \frac{i}{2\lambda^2} \mathbf{1}_{\mathcal{V}_a}. \quad (3.31)$$

This is the continuum matrix obtained from the lower symbol of the finite-lattice time Lax operator itself.

The remaining contribution comes from the overlapping operator products in the discrete zero-curvature equation. For Class 5, denote the differences in Eqs. (2.45) and (2.46) by $\hat{\mathcal{D}}_{5;a,n}^{(\pm)}$. The quadratic coefficients in Eqs. (2.48) and (2.49) contain the products of $A_5^{(1)}$ with

$L_5^{(1)}$; the cubic coefficients contain the products of $A_5^{(2)}$ with $L_5^{(1)}$. Evaluating the shared-site Pauli products with Eq. (2.34) gives

$$\mathcal{D}_{5,2}^{(+)}(\mathbf{S}, \mathbf{S}) = \mathcal{D}_{5,2}^{(-)}(\mathbf{S}, \mathbf{S}) = 2i \left(\frac{\mathbf{1}_{\nu_a}}{4\lambda^2} - U_5^2 \right), \quad (3.32)$$

$$\partial_x S^i \left(\frac{\partial \mathcal{D}_{5,2}^{(+)}}{\partial S_R^i} + \frac{\partial \mathcal{D}_{5,2}^{(-)}}{\partial S_L^i} \right) \Big|_{\mathbf{S}_L = \mathbf{S}_R = \mathbf{S}} = -2i \partial_x (U_5^2), \quad (3.33)$$

$$\mathcal{D}_{5,3}^{(+)}(\mathbf{S}, \mathbf{S}) - \mathcal{D}_{5,3}^{(-)}(\mathbf{S}, \mathbf{S}) = 0. \quad (3.34)$$

The first and third identities use the unit-spin constraint. The second also uses $\mathbf{S}^\top \partial_x \mathbf{S} = 0$. Thus the surviving contribution comes entirely from the first line of Eq. (2.50). In terms of the spatial operator,

$$\begin{aligned} & \widehat{\mathcal{D}}_{5;a,n}^{(+)} - \widehat{\mathcal{D}}_{5;a,n}^{(-)} \\ &= 2i\epsilon^3 \partial_x \left\{ \left[(L_5^{(1)})^2 \right]^\downarrow - \left[(L_5^{(1)})^\downarrow \right]^2 \right\} + \mathcal{O}(\epsilon^4). \end{aligned} \quad (3.35)$$

The algebra in Eq. (3.9) implies

$$(L_5^{(1)})^2 = \frac{1}{4\lambda^2} \mathbf{1}_4, \quad (3.36)$$

and Eq. (3.17) gives $(L_5^{(1)})^\downarrow = U_5$. Hence

$$\left[(L_5^{(1)})^2 \right]^\downarrow - \left[(L_5^{(1)})^\downarrow \right]^2 = \frac{1}{4\lambda^2} \mathbf{1}_{\nu_a} - U_5^2, \quad (3.37)$$

and the continuum contribution defined in Eq. (2.47) is

$$\mathcal{C}_5 = -2i \partial_x (U_5^2). \quad (3.38)$$

Equation (3.38) is absorbed into the time component by choosing

$$\Delta V_5 := -2i U_5^2 + \frac{i}{4\lambda^2} \mathbf{1}_{\nu_a}. \quad (3.39)$$

It obeys

$$[\Delta V_5, U_5] = 0, \quad \partial_x \Delta V_5 + [\Delta V_5, U_5] = \mathcal{C}_5. \quad (3.40)$$

The scalar term in Eq. (3.39) fixes the representative; an x -independent scalar function of λ leaves the curvature unchanged. Combining Eqs. (3.31) and (3.39) gives

$$V_5 = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_5}{\partial S^i} + 2i U_5^2 - \frac{i}{4\lambda^2} \mathbf{1}_{\nu_a}. \quad (3.41)$$

We write $S_x^a := \partial_x S^a$ for $a \in \{+, -, 3\}$. Substituting Eq. (3.18) and using the unit-spin constraint gives the matrix elements

$$\begin{aligned} (V_5)_{11} = \frac{i}{4\lambda^2} & \left[S^3 + \alpha_5 \lambda S^+(1 + S^3) + 2\alpha_5^2 \lambda^2 (S^+)^2 \right. \\ & \left. + \lambda (S^- S_x^+ - S^+ S_x^-) + 4\alpha_5 \lambda^2 (S^+ S_x^3 - S^3 S_x^+) \right], \end{aligned} \quad (3.42)$$

$$\begin{aligned} (V_5)_{12} = \frac{i}{4\lambda^2} & \left[S^- - \alpha_5 \lambda (1 + S^3)^2 - 2\alpha_5^2 \lambda^2 S^+(1 + S^3) \right. \\ & \left. + 2\lambda (S^3 S_x^- - S^- S_x^3) + 2\alpha_5 \lambda^2 (S^+ S_x^- - S^- S_x^+) \right], \end{aligned} \quad (3.43)$$

$$(V_5)_{21} = \frac{i}{4\lambda^2} \left[S^+ + \alpha_5 \lambda (S^+)^2 + 2\lambda (S^+ S_x^3 - S^3 S_x^+) \right], \quad (3.44)$$

$$(V_5)_{22} = \frac{i}{4\lambda^2} \left[-S^3 - \alpha_5 \lambda S^+(1 + S^3) + \lambda (S^+ S_x^- - S^- S_x^+) \right]. \quad (3.45)$$

The difference between Eqs. (3.31) and (3.41) is the contribution required by the overlapping operator products in the finite-lattice zero-curvature equation.

3.3 Zero-curvature equation and the Landau–Lifshitz equations of motion

We now determine the continuum Hamiltonian flow independently of the time Lax matrix derived in Section 3.2. This fixes the dynamical identification left open in Section 2.3 and provides a direct test of the corrected matrix V_5 .

It is convenient to introduce the constant antisymmetric matrix

$$M_5 := \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -i \\ 1 & i & 0 \end{pmatrix}, \quad M_5^\top = -M_5. \quad (3.46)$$

With $\mathbf{S}_n(t) = \mathbf{S}(x_n, t)$ and $\kappa_5 = \epsilon\alpha_5$, the lower symbol of the Class 5 bond Hamiltonian has the long-wavelength expansion

$$h_{5;n,n+1}^\downarrow = 2 + \epsilon^2 \mathcal{H}_5(x_n, t) + \mathcal{O}(\epsilon^3), \quad (3.47)$$

where the Hamiltonian density at (x, t) is

$$\mathcal{H}_5(x, t) = -\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S} + \frac{\alpha_5}{2} \left(\mathbf{S}^\top M_5 \partial_x \mathbf{S} + \partial_x \mathbf{S}^+ \right). \quad (3.48)$$

The last term is a total spatial derivative. With the periodic boundary condition in (2.28), the continuum Hamiltonian obtained from the lattice Hamiltonian is therefore

$$\begin{aligned} H_5^{\text{cl}} &:= \lim_{\epsilon \rightarrow 0} \frac{(H_5^{\text{lat}})^\downarrow - 2N}{\epsilon} \\ &= \int_0^\ell dx \left[-\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S} + \frac{\alpha_5}{2} \mathbf{S}^\top M_5 \partial_x \mathbf{S} \right]. \end{aligned} \quad (3.49)$$

The signs and the overall normalization in (3.49) follow from the lattice Hamiltonian (3.3), the Heisenberg equation (2.18), and the continuum time convention $t = \epsilon^2 t_{\text{lat}}$.

For the unit spin field normalized in (2.28), we use the continuum Poisson bracket

$$\{S^i(x), S^j(y)\}_5 = 2 \varepsilon_{ijk} S^k(x) \delta(x - y), \quad \varepsilon_{123} = 1, \quad (3.50)$$

where $\delta(x - y)$ is the Dirac delta function. The corresponding Hamiltonian flow is

$$\partial_t S^i(x) = \{S^i(x), H_5^{\text{cl}}\}_5. \quad (3.51)$$

Using the bilinear cross product $(\mathbf{A} \times \mathbf{B})^i = \varepsilon_{ijk} A^j B^k$, the equations of motion take the compact form

$$\partial_t \mathbf{S} = -2 \mathbf{S} \times \left(\partial_x^2 \mathbf{S} + \alpha_5 M_5 \partial_x \mathbf{S} \right). \quad (3.52)$$

The right-hand side is tangent to the unit-spin constraint, so $\partial_t(\mathbf{S}^\top \mathbf{S}) = 0$.

For comparison with the matrix entries of the Lax pair, the same equations can be written in terms of the transverse components $S^\pm$ defined in (3.14). Using $\mathbf{S}^\top \partial_x \mathbf{S} = 0$, one obtains

$$\partial_t S^+ = 2i \left(S^+ \partial_x^2 S^3 - S^3 \partial_x^2 S^+ + \alpha_5 S^+ \partial_x S^+ \right), \quad (3.53)$$

$$\partial_t S^- = 2i \left(S^3 \partial_x^2 S^- - S^- \partial_x^2 S^3 + \alpha_5 S^+ \partial_x S^- \right), \quad (3.54)$$

$$\partial_t S^3 = i \left(S^- \partial_x^2 S^+ - S^+ \partial_x^2 S^- + 2\alpha_5 S^+ \partial_x S^3 \right). \quad (3.55)$$

A direct expansion of the lower symbol of the lattice Heisenberg derivative then gives

$$\left(\partial_{t_{\text{lat}}} \widehat{L}_{5;a,n} \right)^\downarrow = \epsilon^3 \partial_t U_5(x_n, t; \lambda) + \mathcal{O}(\epsilon^4), \quad (3.56)$$

with $\partial_t \mathbf{S}$ given by (3.52). Equation (3.56) therefore supplies the dynamical identification required on the left-hand side of the continuum equation (2.51).

To establish the off-shell relation between the curvature and the equations of motion, we treat the time derivatives as independent variables and define the three equation-of-motion residuals

$$E_+^{(5)} := \partial_t S^+ - 2i \left(S^+ \partial_x^2 S^3 - S^3 \partial_x^2 S^+ + \alpha_5 S^+ \partial_x S^+ \right), \quad (3.57)$$

$$E_-^{(5)} := \partial_t S^- - 2i \left(S^3 \partial_x^2 S^- - S^- \partial_x^2 S^3 + \alpha_5 S^+ \partial_x S^- \right), \quad (3.58)$$

$$E_3^{(5)} := \partial_t S^3 - i \left(S^- \partial_x^2 S^+ - S^+ \partial_x^2 S^- + 2\alpha_5 S^+ \partial_x S^3 \right). \quad (3.59)$$

The curvature of the matrices U_5 and V_5 in (3.18) and (3.41) is

$$F_{tx}^{(5)} := \partial_t U_5 - \partial_x V_5 + [U_5, V_5]. \quad (3.60)$$

Using only the unit-spin constraint and its spatial derivatives, the curvature factorizes as

$$F_{tx}^{(5)} = \frac{1}{4\lambda} \begin{pmatrix} E_3^{(5)} + 2\alpha_5 \lambda E_+^{(5)} & E_-^{(5)} - 2\alpha_5 \lambda E_3^{(5)} \\ E_+^{(5)} & -E_3^{(5)} \end{pmatrix}. \quad (3.61)$$

For $\lambda \neq 0$, the residuals are recovered from the curvature entries by

$$\begin{aligned} E_+^{(5)} &= 4\lambda (F_{tx}^{(5)})_{21}, & E_3^{(5)} &= -4\lambda (F_{tx}^{(5)})_{22}, \\ E_-^{(5)} &= 4\lambda \left((F_{tx}^{(5)})_{12} - 2\alpha_5 \lambda (F_{tx}^{(5)})_{22} \right). \end{aligned} \quad (3.62)$$

Hence the zero-curvature condition $F_{tx}^{(5)} = 0$ is equivalent to the three continuum equations of motion (3.53)–(3.55).

To compare with the direct continuum construction of Ref. [4], we translate our conventions by

$$S^2 \mapsto -S^2, \quad \alpha_5 \mapsto -\alpha_5, \quad t \mapsto 2t. \quad (3.63)$$

With these substitutions, Eq. (3.52) takes the form of the Class 5 continuum equation used in Ref. [4]. The reflection of the second spin axis conjugates M_5 by $\text{diag}(1, -1, 1)$, yielding the deformation matrix used there.

3.4 Comparison with the classical r -matrix construction

The time Lax matrix in Eq. (3.41) was obtained by starting from the finite-lattice time operator. We now construct the time component independently from the classical r -matrix and monodromy of the continuum model.

To keep the two roles of the spectral parameter separate, let $\zeta \in \mathbb{C}$ denote the *generating spectral parameter* used to expand the transfer matrix, while λ continues to denote the Lax spectral parameter of the final pair (U_5, V_5) . Define the rank-one coherent-state projector

$$\rho(x, t) := \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad \rho^2 = \rho, \quad \text{tr } \rho = 1, \quad (3.64)$$

and the deformation matrix in the auxiliary space

$$K_5^{\text{cl}}(x, t) := \frac{1}{2} \begin{pmatrix} S^+ & -(1 + S^3) \\ 0 & 0 \end{pmatrix}. \quad (3.65)$$

These are precisely the lower symbols in Eqs. (3.15) and (3.16). The spatial Lax matrix therefore admits the form

$$U_5(x, t; \lambda) = \frac{\rho(x, t)}{2\lambda} + \alpha_5 K_5^{\text{cl}}(x, t). \quad (3.66)$$

We suppress the common time argument in the following formulas and first fix the classical r -matrix in the conventions of this paper. Let P_{12} be the permutation operator on two copies of the two-dimensional auxiliary space. On either copy, define the projector $P_\uparrow$ and the matrix unit E_{12} by

$$P_\uparrow := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E_{12} := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (3.67)$$

and the constant Jordanian tensor

$$C_{12} := P_\uparrow \otimes E_{12} - E_{12} \otimes P_\uparrow. \quad (3.68)$$

A direct calculation with the spin Poisson bracket (3.50) gives

$$r_{12}^{(5)}(\zeta, \lambda) = \frac{i}{2(\zeta - \lambda)} P_{12} + i\alpha_5 C_{12}, \quad (3.69)$$

for which

$$\{U_{5,1}(x; \zeta), U_{5,2}(y; \lambda)\}_5 = [r_{12}^{(5)}(\zeta, \lambda), U_{5,1}(x; \zeta) + U_{5,2}(y; \lambda)] \delta(x - y). \quad (3.70)$$

Here $U_{5,1} := U_5 \otimes \mathbf{1}_2$ and $U_{5,2} := \mathbf{1}_2 \otimes U_5$ act on the first and second auxiliary spaces, respectively. The rational part of Eq. (3.69) is the usual isotropic Landau–Lifshitz contribution, whereas the constant term contains the Class 5 deformation. This structure is consistent with the classical r -matrix obtained from the Class 5 quantum model in Ref. [4].

Let $\mathcal{P}$ denote path ordering with larger spatial coordinates to the left. The classical monodromy matrix is

$$T(x, y; \zeta) := \mathcal{P} \exp \left(\int_y^x U_5(s; \zeta) ds \right). \quad (3.71)$$

The associated transfer matrix on the periodic interval of length ℓ is

$$\tau(\zeta) := \text{tr } T(\ell, 0; \zeta). \quad (3.72)$$

For a linear classical r -matrix algebra of the form (3.70), the standard monodromy construction gives the generating time matrix

$$\mathbb{V}_2(x; \zeta, \lambda) = \frac{1}{\tau(\zeta)} \text{tr}_1 \left[T_1(\ell, x; \zeta) r_{12}^{(5)}(\zeta, \lambda) T_1(x, 0; \zeta) \right], \quad (3.73)$$

where the trace is taken over the first auxiliary space [14, 6].

The pole of $U_5(x; \zeta)$ at $\zeta = 0$ has residue $\rho/2$, by Eq. (3.66). The local expansion of the monodromy is therefore governed by a rank-one projector $\Pi(x; \zeta)$ satisfying

$$\Pi^2 = \Pi, \quad \text{tr } \Pi = 1, \quad \partial_x \Pi = [U_5(x; \zeta), \Pi], \quad (3.74)$$

with the branch

$$\Pi(x; \zeta) = \rho(x) + \zeta \Pi_1(x) + \mathcal{O}(\zeta^2). \quad (3.75)$$

Expanding Eq. (3.74) to first order fixes the coefficient uniquely as

$$\Pi_1 = 2[\rho, \partial_x \rho] - 2\alpha_5[\rho, [K_5^{\text{cl}}, \rho]]. \quad (3.76)$$

Using Eq. (3.75) in the generating matrix (3.73), write

$$\mathbb{V}_2(x; \zeta, \lambda) = \mathbb{V}^{(0)}(x; \lambda) + \zeta \mathbb{V}^{(1)}(x; \lambda) + \mathcal{O}(\zeta^2). \quad (3.77)$$

The coefficient relevant for the Class 5 Hamiltonian flow is

$$\begin{aligned} \mathbb{V}^{(1)} = & -\frac{i}{2\lambda} \left(\Pi_1 + \frac{\rho}{\lambda} \right) \\ & + i\alpha_5 [\text{tr}(\Pi_1 P_\uparrow) E_{12} - \text{tr}(\Pi_1 E_{12}) P_\uparrow]. \end{aligned} \quad (3.78)$$

The standard monodromy formula is conventionally written with the Hamiltonian in the first argument of the Poisson bracket. Our Hamiltonian evolution in Eq. (3.51) uses the opposite order. With this convention, the time Lax matrix obtained from the classical r -matrix route is therefore

$$V_{5,r}(x; \lambda) := -\mathbb{V}^{(1)}(x; \lambda). \quad (3.79)$$

Substituting Eq. (3.76) into Eq. (3.78) and using only the unit spin constraint gives

$$V_{5,r}(x; \lambda) - V_5(x; \lambda) = \frac{i}{4\lambda^2} \mathbf{1}_2. \quad (3.80)$$

The matrix on the right-hand side is independent of the spin field and of the spatial coordinate. Its spatial derivative vanishes and it commutes with U_5 , so $V_{5,r}$ and V_5 generate the same zero-curvature equation. This is the scalar freedom described after Eq. (3.39).

4 Class 6 model

We now apply the finite-lattice construction to the Class 6 chain, using the quantum-chain normalization of Appendix A of Ref. [4]. The spatial expansion contains a nonzero second-order operator coefficient. We determine its contribution to the time Lax matrix, including the commutator term in the lower-symbol correction, and then establish the off-shell equivalence of the corrected Lax equation and the Hamiltonian equations of motion.

4.1 Quantum Class 6 model and spatial Lax matrix

Let $\kappa_6 \in \mathbb{C}$ denote the *Class 6 lattice deformation parameter*. We use the ordered two-spin basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ and fix the coefficient of the permutation interaction to be 2. In this normalization the local Hamiltonian is

$$h_6(\kappa_6) = \begin{pmatrix} 2 & \kappa_6 & \kappa_6 & 0 \\ 0 & 0 & 2 & -\kappa_6 \\ 0 & 2 & 0 & -\kappa_6 \\ 0 & 0 & 0 & 2 \end{pmatrix} = 2P + \kappa_6 K_6, \quad (4.1)$$

where P is the permutation operator in Eq. (2.9), and the two-site deformation operator K_6 is defined by

$$K_6 := \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (4.2)$$

Its algebra with P is

$$PK_6 = K_6P = K_6, \quad K_6^3 = 0, \quad K_6^2 = \begin{pmatrix} 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (4.3)$$

For the periodic chain of N sites, the quantum Hamiltonian is

$$H_6^{\text{lat}}(\kappa_6) = \sum_{n=1}^N h_{6;n,n+1}(\kappa_6), \quad h_{6;N,N+1} \equiv h_{6;N,1}. \quad (4.4)$$

The Hamiltonian is non-Hermitian for $\kappa_6 \neq 0$ in the standard inner product. At $\kappa_6 = 0$ it reduces to the spin- $\frac{1}{2}$ XXX chain with the same exchange normalization as in Section 3.1.

Using the quantum spectral parameter u of Section 2.1, the regular R -matrix generating Eq. (4.1) is

$$R_6(u; \kappa_6) = 2u\mathbf{1}_4 + P + \kappa_6 u(1+2u)K_6 + \frac{\kappa_6^2 u^2 (1+2u)^2}{2} K_6^2 \quad (4.5)$$

$$= \begin{pmatrix} 1+2u & \kappa_6 u(1+2u) & \kappa_6 u(1+2u) & -\kappa_6^2 u^2 (1+2u)^2 \\ 0 & 2u & 1 & -\kappa_6 u(1+2u) \\ 0 & 1 & 2u & -\kappa_6 u(1+2u) \\ 0 & 0 & 0 & 1+2u \end{pmatrix}. \quad (4.6)$$

This is Eq. (A.1) of Ref. [4] with the exchange normalization fixed above. It satisfies the Yang–Baxter equation and

$$R_6(0; \kappa_6) = P, \quad P \left. \frac{\partial R_6(u; \kappa_6)}{\partial u} \right|_{u=0} = h_6(\kappa_6). \quad (4.7)$$

All quantum spectral derivatives are taken at fixed κ_6 . The identities in Eq. (4.3) also give

$$R_6(u; \kappa_6) R_6(-u; \kappa_6) = (1 - 4u^2) \mathbf{1}_4. \quad (4.8)$$

This algebraic inversion relation ensures that the inverse required by the finite-lattice time operator exists for $u \neq \pm \frac{1}{2}$.

To fix the continuum scaling, consider the lower symbol of a bond Hamiltonian. The coherent states and the smooth unit spin field $\mathbf{S}(x, t)$ are those of Section 2.2. We write the transverse components as

$$S^\pm := S^1 \pm iS^2, \quad S_n^\pm := S^\pm(x_n, t). \quad (4.9)$$

With the complex extension described in Section 2.2, the unit-spin constraint reads $S^+ S^- + (S^3)^2 = 1$. Direct evaluation of Eq. (4.1) gives

$$h_{6;n,n+1}^\downarrow = 1 + \mathbf{S}_n^\top \mathbf{S}_{n+1} + \frac{\kappa_6}{2} (S_n^+ S_{n+1}^3 + S_n^3 S_{n+1}^+). \quad (4.10)$$

After subtracting the constant 2, the exchange term starts at order ϵ^2 by Eq. (2.30), whereas the deformation term tends to $\kappa_6 S^+ S^3$ for coincident spins. Both terms therefore survive in the same continuum limit if the *Class 6 continuum deformation coupling* α_6 and the continuum spectral parameter $\lambda \neq 0$ are held fixed under

$$\kappa_6 = \epsilon^2 \alpha_6, \quad u = \frac{\lambda}{\epsilon}, \quad t = \epsilon^2 t_{\text{lat}}, \quad \lambda \neq 0. \quad (4.11)$$

Here ϵ is the lattice spacing, u and λ are the quantum and continuum spectral parameters, and t_{lat} and t are the lattice and continuum times. The deformation scaling agrees with Appendix A.1 of Ref. [4]; the spectral scaling ensures that the normalized spatial Lax operator approaches the identity. If lattice parameters are dimensionless, λ has the dimension of length and α_6 has the dimension of inverse length squared.

In the fundamental representation, define $L_{6;a,n}(u) := R_{6;a,n}(u; \kappa_6)$ on $\mathcal{V}_a \otimes \mathcal{H}_n$ and normalize it as

$$\widehat{L}_{6;a,n}(\epsilon, \lambda) := \frac{1}{2u} L_{6;a,n}(u). \quad (4.12)$$

Substituting Eq. (4.11) into Eq. (4.5) yields the exact finite expansion

$$\widehat{L}_{6;a,n} = \mathbf{1}_{\mathcal{V}_a \otimes \mathcal{H}_n} + \epsilon L_{6;a,n}^{(1)} + \epsilon^2 L_{6;a,n}^{(2)} + \epsilon^3 L_{6;a,n}^{(3)}. \quad (4.13)$$

The superscript labels the power of ϵ , and each coefficient is the copy on $\mathcal{V}_a \otimes \mathcal{H}_n$ of the two-site operator

$$L_6^{(1)} = \frac{P}{2\lambda} + \alpha_6 \lambda K_6 + \alpha_6^2 \lambda^3 K_6^2, \quad (4.14)$$

$$L_6^{(2)} = \frac{\alpha_6}{2} K_6 + \alpha_6^2 \lambda^2 K_6^2, \quad L_6^{(3)} = \frac{\alpha_6^2 \lambda}{4} K_6^2. \quad (4.15)$$

In particular, the operator square of the first coefficient satisfies

$$(L_6^{(1)})^2 = \frac{\mathbf{1}_4}{4\lambda^2} + 2L_6^{(2)}. \quad (4.16)$$

The second coefficient is therefore already present in products of the leading spatial coefficients. It will enter the coherent-state product correction in the time Lax equation.

For the lower-symbol evaluation at $x = x_n$, the permutation gives

$$P_{a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad (4.17)$$

while the deformation operators give

$$K_{6;a,n}^\downarrow = \begin{pmatrix} S^+/2 & S^3 \\ 0 & -S^+/2 \end{pmatrix}, \quad (K_{6;a,n}^2)^\downarrow = \begin{pmatrix} 0 & -S^+ \\ 0 & 0 \end{pmatrix}. \quad (4.18)$$

The second matrix is obtained by squaring the quantum operator before taking its lower symbol. Using these matrices in Eq. (4.14), the continuum spatial Lax matrix is

$$\begin{aligned} U_6(x, t; \lambda) &:= (L_{6;a,n}^{(1)})^\downarrow|_{x_n=x} \\ &= \frac{1}{4\lambda} \begin{pmatrix} 1 + S^3 + 2\alpha_6 \lambda^2 S^+ & S^- + 4\alpha_6 \lambda^2 S^3 - 4\alpha_6^2 \lambda^4 S^+ \\ S^+ & 1 - S^3 - 2\alpha_6 \lambda^2 S^+ \end{pmatrix}. \end{aligned} \quad (4.19)$$

It obeys

$$\widehat{L}_{6;a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U_6(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad \text{tr } U_6 = \frac{1}{2\lambda}. \quad (4.20)$$

We retain the complete $\mathfrak{gl}(2)$ matrix in the finite-lattice limit and denote its traceless part by

$$U_{6,0} := U_6 - \frac{1}{4\lambda} \mathbf{1}_2. \quad (4.21)$$

At $\alpha_6 = 0$, Eq. (4.19) reduces to $(\mathbf{1}_2 + S^i \sigma^i)/(4\lambda)$, the spatial Lax matrix of the isotropic Landau–Lifshitz model in the normalization used here. For the deformed model, Eqs. (4.13)–(4.16) provide the operator coefficients needed to construct the time component.

4.2 Time Lax matrix from the finite-lattice construction

We apply the finite-lattice construction of Eq. (2.19) to the Class 6 operators of Section 4.1. As in the Class 5 calculation, we remove a field-independent scalar term by choosing

$$\hat{A}_{6;a,n} := A_{6;a,n} + \frac{i}{u} \mathbf{1}, \quad (4.22)$$

where the identity acts on the auxiliary space and the physical sites $n-1, n$. The normalized spatial operator then gives

$$\hat{A}_{6;a,n} = -i\hat{L}_{6;a,n}^{-1} \left([h_{6;n-1,n}, \hat{L}_{6;a,n}] + \partial_u \hat{L}_{6;a,n} \right). \quad (4.23)$$

The derivative is taken at fixed lattice coupling κ_6 . With the scaling in Eq. (4.11), the exact expansion (4.13) therefore implies

$$\partial_u \hat{L}_{6;a,n} = \epsilon^2 \partial_\lambda L_{6;a,n}^{(1)} + \epsilon^3 \partial_\lambda L_{6;a,n}^{(2)} + \epsilon^4 \partial_\lambda L_{6;a,n}^{(3)}. \quad (4.24)$$

Here and below, ∂_λ holds α_6 fixed, and also holds the spin fixed when acting on a classical matrix.

The bond Hamiltonian is

$$h_{6;n-1,n} = h_{n-1,n}^{(0)} + \epsilon^2 \alpha_6 K_{6;n-1,n}, \quad h^{(0)} := 2P. \quad (4.25)$$

Expanding Eq. (4.23) gives

$$\hat{A}_{6;a,n} = \epsilon A_{6;a,n}^{(1)} + \epsilon^2 A_{6;a,n}^{(2)} + \mathcal{O}(\epsilon^3), \quad (4.26)$$

where

$$A_{6;a,n}^{(1)} = -i[h_{n-1,n}^{(0)}, L_{6;a,n}^{(1)}], \quad (4.27)$$

$$\begin{aligned} A_{6;a,n}^{(2)} &= iL_{6;a,n}^{(1)}[h_{n-1,n}^{(0)}, L_{6;a,n}^{(1)}] \\ &\quad - i[h_{n-1,n}^{(0)}, L_{6;a,n}^{(2)}] - i\partial_\lambda L_{6;a,n}^{(1)}. \end{aligned} \quad (4.28)$$

The deformation of the bond Hamiltonian first contributes explicitly at order ϵ^3 , since its commutator with the leading identity in $\hat{L}_6$ vanishes. The dependence on α_6 in Eqs. (4.27) and (4.28) enters through $L_6^{(1)}$ and $L_6^{(2)}$.

Taking the lower symbol on the two physical sites gives

$$(A_{6;a,n}^{(1)})^\downarrow = -2(\mathbf{S}_{n-1} \times \mathbf{S}_n)_i \frac{\partial U_6}{\partial S^i}. \quad (4.29)$$

The Cartesian derivatives in this formula act on the explicit matrix (4.19), with $S^\pm = S^1 \pm iS^2$, before imposing the unit-spin constraint. Since U_6 is affine in the spin components, these derivatives are independent of the spin. At $x = x_n$,

$$\mathbf{S}_{n-1} \times \mathbf{S}_n = \epsilon \mathbf{S} \times \partial_x \mathbf{S} + \mathcal{O}(\epsilon^2). \quad (4.30)$$

The second time coefficient is needed only at coincident spin arguments. Using $S^+ S^- + (S^3)^2 = 1$ and the traceless spatial matrix $U_{6,0}$ of Eq. (4.21), we obtain

$$(A_{6;a,n}^{(2)})^\downarrow \Big|_{\mathbf{S}_{n-1}=\mathbf{S}_n=\mathbf{S}} = -2i\partial_\lambda U_{6,0} + i\alpha_6 S^+ S^3 \mathbf{1}_{\mathcal{V}_a}. \quad (4.31)$$

Consequently,

$$\hat{A}_{6;a,n}^\downarrow = \epsilon^2 V_6^\downarrow(x_n, t; \lambda) + \mathcal{O}(\epsilon^3), \quad (4.32)$$

with

$$V_6^\downarrow = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_6}{\partial S^i} - 2i\partial_\lambda U_{6,0} + i\alpha_6 S^+ S^3 \mathbf{1}_{\mathcal{V}_a}. \quad (4.33)$$

The products in the discrete zero-curvature equation require a separate calculation. Let $\widehat{\mathcal{D}}_{6;a,n}^{(\pm)}$ denote Eqs. (2.45) and (2.46) evaluated for Class 6. For the first product, the operator expansions give

$$\begin{aligned} \widehat{\mathcal{D}}_{6;a,n}^{(+)} &= \epsilon^2 \left[(A_{6;a,n+1}^{(1)} L_{6;a,n}^{(1)})^\downarrow - (A_{6;a,n+1}^{(1)})^\downarrow (L_{6;a,n}^{(1)})^\downarrow \right] \\ &\quad + \epsilon^3 \left[(A_{6;a,n+1}^{(2)} L_{6;a,n}^{(1)} + A_{6;a,n+1}^{(1)} L_{6;a,n}^{(2)})^\downarrow \right. \\ &\quad \left. - (A_{6;a,n+1}^{(2)})^\downarrow (L_{6;a,n}^{(1)})^\downarrow - (A_{6;a,n+1}^{(1)})^\downarrow (L_{6;a,n}^{(2)})^\downarrow \right] + \mathcal{O}(\epsilon^4). \end{aligned} \quad (4.34)$$

The expression for $\widehat{\mathcal{D}}_{6;a,n}^{(-)}$ has the spatial factor on the left and the time factor at n , supported on $n-1, n$. Terms involving the third time coefficient cancel within each difference, because they multiply the leading identity in $\widehat{L}_6$. The coefficient $L_6^{(3)}$ first enters these differences at order ϵ^4 . The continuum source therefore requires the products displayed in Eq. (4.34), including those involving $L_6^{(2)}$.

At coincident spins, the order- ϵ^2 terms cancel between $\widehat{\mathcal{D}}_{6;a,n}^{(+)}$ and $\widehat{\mathcal{D}}_{6;a,n}^{(-)}$. To express the surviving contribution, define Ξ_6 as the difference between the lower symbol of the square of the first spatial coefficient and the square of its lower symbol:

$$\begin{aligned} \Xi_6 &:= \left[(L_6^{(1)})^2 \right]^\downarrow - U_6^2 \\ &= \frac{1}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a} + 2(L_6^{(2)})^\downarrow - U_6^2. \end{aligned} \quad (4.35)$$

The second equality follows from Eq. (4.16). In particular, Eq. (4.18) gives

$$(L_6^{(2)})^\downarrow = \begin{pmatrix} \alpha_6 S^+ / 4 & \alpha_6 S^3 / 2 - \alpha_6^2 \lambda^2 S^+ \\ 0 & -\alpha_6 S^+ / 4 \end{pmatrix}. \quad (4.36)$$

The first bracket in Eq. (4.34) is $\mathcal{D}_{6,2}^{(+)}$, and the second is $\mathcal{D}_{6,3}^{(-)}$, with the two spin arguments understood as in Eq. (2.48). The corresponding left-ordered products give $\mathcal{D}_{6,2}^{(-)}$ and $\mathcal{D}_{6,3}^{(-)}$. Applying Eq. (2.34) at the shared physical site yields

$$\mathcal{D}_{6,2}^{(+)}(\mathbf{S}, \mathbf{S}) = \mathcal{D}_{6,2}^{(-)}(\mathbf{S}, \mathbf{S}) = 2i\Xi_6, \quad (4.37)$$

$$\partial_x S^i \left(\frac{\partial \mathcal{D}_{6,2}^{(+)}}{\partial S_R^i} + \frac{\partial \mathcal{D}_{6,2}^{(-)}}{\partial S_L^i} \right) \Big|_{S_L=S_R=\mathbf{S}} = 2i\partial_x \Xi_6, \quad (4.38)$$

$$\mathcal{D}_{6,3}^{(+)}(\mathbf{S}, \mathbf{S}) - \mathcal{D}_{6,3}^{(-)}(\mathbf{S}, \mathbf{S}) = 2i[\Xi_6, U_6]. \quad (4.39)$$

The first identity cancels the order- ϵ^2 difference. The remaining identities display separately the two contributions in Eq. (2.50): the spatial variation supplies the derivative, and the cubic products at coincident spins supply the commutator. Consequently,

$$\mathcal{C}_6 := \lim_{\epsilon \rightarrow 0} \epsilon^{-3} \left(\widehat{\mathcal{D}}_{6;a,n}^{(+)} - \widehat{\mathcal{D}}_{6;a,n}^{(-)} \right) = 2i(\partial_x \Xi_6 + [\Xi_6, U_6]). \quad (4.40)$$

This calculation uses the unit-spin constraint and its spatial derivative and holds for arbitrary time dependence of the spin.

The local time correction

$$\begin{aligned}\Delta V_6 &:= 2i\Xi_6 - \frac{i}{4\lambda^2}\mathbf{1}_{\mathcal{V}_a} \\ &= 4i(L_6^{(2)})^\downarrow - 2iU_6^2 + \frac{i}{4\lambda^2}\mathbf{1}_{\mathcal{V}_a}\end{aligned}\quad (4.41)$$

then satisfies

$$\partial_x \Delta V_6 + [\Delta V_6, U_6] = \mathcal{C}_6. \quad (4.42)$$

The field-independent scalar in Eq. (4.41) fixes a representative with traceless final time matrix. To simplify it, the explicit spatial matrix gives

$$\begin{aligned}\det U_6 &= -\frac{\alpha_6}{2}S^+S^3, \\ \partial_\lambda U_{6,0} &= 4(L_6^{(2)})^\downarrow - \frac{1}{\lambda}U_{6,0}.\end{aligned}\quad (4.43)$$

The determinant identity uses the unit-spin constraint. The Cayley–Hamilton identity, together with $\text{tr } U_6 = 1/(2\lambda)$, now reduces Eq. (4.41) to

$$\Delta V_6 = i\partial_\lambda U_{6,0} - i\alpha_6 S^+ S^3 \mathbf{1}_{\mathcal{V}_a}. \quad (4.44)$$

Adding this correction to Eq. (4.33) yields

$$V_6 := V_6^\downarrow + \Delta V_6 = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_6}{\partial S^i} - i\partial_\lambda U_{6,0}, \quad \text{tr } V_6 = 0. \quad (4.45)$$

We write $S_x^a := \partial_x S^a$ for $a \in \{+, -, 3\}$. The matrix elements are

$$\begin{aligned}(V_6)_{11} &= \frac{i}{4\lambda^2} \left[S^3 - 2\alpha_6 \lambda^2 S^+ + \lambda(S^- S_x^+ - S^+ S_x^-) \right. \\ &\quad \left. + 4\alpha_6 \lambda^3 (S^+ S_x^3 - S^3 S_x^+) \right],\end{aligned}\quad (4.46)$$

$$\begin{aligned}(V_6)_{12} &= \frac{i}{4\lambda^2} \left[S^- - 4\alpha_6 \lambda^2 S^3 + 12\alpha_6^2 \lambda^4 S^+ \right. \\ &\quad \left. + 2\lambda(S^3 S_x^- - S^- S_x^3) + 4\alpha_6 \lambda^3 (S^- S_x^+ - S^+ S_x^-) \right. \\ &\quad \left. + 8\alpha_6^2 \lambda^5 (S^3 S_x^+ - S^+ S_x^3) \right],\end{aligned}\quad (4.47)$$

$$(V_6)_{21} = \frac{i}{4\lambda^2} \left[S^+ + 2\lambda(S^+ S_x^3 - S^3 S_x^+) \right], \quad (4.48)$$

$$(V_6)_{22} = -(V_6)_{11}. \quad (4.49)$$

The role of the second spatial coefficient is also visible directly in the source. Expanding Eq. (4.40) gives

$$\mathcal{C}_6 = -2i\partial_x(U_6^2) + 4i\partial_x(L_6^{(2)})^\downarrow + 4i \left[(L_6^{(2)})^\downarrow, U_6 \right]. \quad (4.50)$$

At a fixed time, consider the spatially uniform spin configuration $\mathbf{S} = (0, 0, 1)^\top$. Then

$$U_6 = \begin{pmatrix} 1/(2\lambda) & \alpha_6 \lambda \\ 0 & 0 \end{pmatrix}, \quad \mathcal{C}_6 = \begin{pmatrix} 0 & -i\alpha_6/\lambda \\ 0 & 0 \end{pmatrix}. \quad (4.51)$$

The source is nonzero for $\alpha_6 \neq 0$. Omitting the term involving $L_6^{(2)}$ from Eq. (4.41) would leave a polynomial in U_6 , whose spatial derivative and commutator with U_6 both vanish in this example. The total-derivative correction sufficient for Class 5 therefore fails for Class 6.

In Section 4.3 we derive the Hamiltonian equations independently and establish their off-shell equivalence to the zero-curvature equation for (U_6, V_6) .

4.3 Zero-curvature equation and the Landau–Lifshitz equations of motion

We derive the continuum Hamiltonian flow from the Class 6 bond Hamiltonian and compare it with the corrected Lax pair of Section 4.2. The Hamiltonian also fixes the continuum limit of the Heisenberg derivative on the left-hand side of the lattice zero-curvature equation.

With $\mathbf{S}_n(t) = \mathbf{S}(x_n, t)$ and $\kappa_6 = \epsilon^2 \alpha_6$, expanding the bond lower symbol in Eq. (4.10) gives

$$h_{6;n,n+1}^\downarrow = 2 + \epsilon^2 \mathcal{H}_6(x_n, t) + \mathcal{O}(\epsilon^3), \quad (4.52)$$

where the continuum Hamiltonian density at (x, t) is

$$\mathcal{H}_6(x, t) = -\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S} + \alpha_6 S^+ S^3. \quad (4.53)$$

Here we used the spatial derivatives of the unit-spin constraint in Eq. (2.30). The anisotropy can be written in terms of the constant symmetric matrix

$$N_6 := \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -i \\ -1 & -i & 0 \end{pmatrix}, \quad N_6^\top = N_6, \quad N_6^3 = 0. \quad (4.54)$$

Since $\mathbf{S}^\top N_6 \mathbf{S} = -2S^+ S^3$, the continuum Hamiltonian at fixed circumference $\ell = N\epsilon$ is

$$\begin{aligned} H_6^{\text{cl}} &:= \lim_{\epsilon \rightarrow 0} \frac{(H_6^{\text{lat}})^\downarrow - 2N}{\epsilon} \\ &= \int_0^\ell dx \left[-\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S} - \frac{\alpha_6}{2} \mathbf{S}^\top N_6 \mathbf{S} \right]. \end{aligned} \quad (4.55)$$

We use the equal-time spin Poisson bracket with the same normalization as in Eq. (3.50),

$$\{S^i(x), S^j(y)\}_6 = 2 \varepsilon_{ijk} S^k(x) \delta(x - y), \quad \varepsilon_{123} = 1, \quad (4.56)$$

where ε_{ijk} is completely antisymmetric and $\delta(x - y)$ is the Dirac delta function. The common time argument is suppressed in Eq. (4.56). We take the functional derivative with respect to the Cartesian spin components before imposing $\mathbf{S}^\top \mathbf{S} = 1$. Integration by parts with periodic variations then gives $\delta H_6^{\text{cl}} / \delta \mathbf{S} = \partial_x^2 \mathbf{S} - \alpha_6 N_6 \mathbf{S}$. The Hamiltonian equation $\partial_t S^i = \{S^i, H_6^{\text{cl}}\}_6$ therefore reads

$$\partial_t \mathbf{S} = -2 \mathbf{S} \times \left(\partial_x^2 \mathbf{S} - \alpha_6 N_6 \mathbf{S} \right). \quad (4.57)$$

The cross product is bilinear, as in Section 3.3, and the right-hand side preserves $\mathbf{S}^\top \mathbf{S} = 1$. In terms of $S^\pm$ and S^3 , Eq. (4.57) becomes

$$\partial_t S^+ = 2i \left(S^+ \partial_x^2 S^3 - S^3 \partial_x^2 S^+ + \alpha_6 (S^+)^2 \right), \quad (4.58)$$

$$\partial_t S^- = 2i \left(S^3 \partial_x^2 S^- - S^- \partial_x^2 S^3 + 2\alpha_6 (S^3)^2 - \alpha_6 S^+ S^- \right), \quad (4.59)$$

$$\partial_t S^3 = i \left(S^- \partial_x^2 S^+ - S^+ \partial_x^2 S^- - 2\alpha_6 S^+ S^3 \right). \quad (4.60)$$

The relation to the quantum time evolution can also be checked at finite lattice spacing. Let σ_n^i denote the Pauli matrix σ^i acting at site n , with identity operators on the other sites, and write $\boldsymbol{\sigma}_n = (\sigma_n^1, \sigma_n^2, \sigma_n^3)^\top$. Taking the lower symbol of the Heisenberg derivative componentwise on a product of coherent states at a fixed lattice time gives

$$\begin{aligned} \frac{1}{\epsilon^2} (\partial_{t_{\text{lat}}} \boldsymbol{\sigma}_n)^\downarrow &= -2\mathbf{S}_n \times \left[\frac{\mathbf{S}_{n-1} + \mathbf{S}_{n+1} - 2\mathbf{S}_n}{\epsilon^2} \right. \\ &\quad \left. - \frac{\alpha_6}{2} N_6 (\mathbf{S}_{n-1} + \mathbf{S}_{n+1}) \right]. \end{aligned} \quad (4.61)$$

This identity follows from the two bonds adjacent to site n and the Pauli commutation relations. Its smooth-field limit is Eq. (4.57), with the time convention $t = \epsilon^2 t_{\text{lat}}$. Applying the same commutator to the normalized spatial operator gives

$$\left(\partial_{t_{\text{lat}}} \widehat{L}_{6;a,n}\right)^\downarrow = \epsilon^3 \partial_t U_6(x_n, t; \lambda) + \mathcal{O}(\epsilon^4), \quad (4.62)$$

where $\partial_t \mathbf{S}$ is given by Eq. (4.57). Together with the correction identity (4.42), this supplies both sides of the continuum zero-curvature equation.

To establish equivalence with the equations of motion, we now treat the time derivatives as independent variables. Define the equation-of-motion residual vector

$$\mathbf{E}^{(6)} := \partial_t \mathbf{S} + 2\mathbf{S} \times \left(\partial_x^2 \mathbf{S} - \alpha_6 N_6 \mathbf{S}\right), \quad (4.63)$$

Write $E_i^{(6)}$ for its Cartesian components and $E_\pm^{(6)} := E_1^{(6)} \pm iE_2^{(6)}$ for its transverse components. The curvature of the matrices in Eqs. (4.19) and (4.45) is

$$F_{tx}^{(6)} := \partial_t U_6 - \partial_x V_6 + [U_6, V_6]. \quad (4.64)$$

Using only $\mathbf{S}^\top \mathbf{S} = 1$, $\mathbf{S}^\top \partial_x \mathbf{S} = 0$, and $\mathbf{S}^\top \partial_x^2 \mathbf{S} = -\partial_x \mathbf{S}^\top \partial_x \mathbf{S}$, we obtain the off-shell identity

$$F_{tx}^{(6)} = \frac{1}{4\lambda} \begin{pmatrix} E_3^{(6)} + 2\alpha_6 \lambda^2 E_+^{(6)} & E_-^{(6)} + 4\alpha_6 \lambda^2 E_3^{(6)} - 4\alpha_6^2 \lambda^4 E_+^{(6)} \\ E_+^{(6)} & -E_3^{(6)} - 2\alpha_6 \lambda^2 E_+^{(6)} \end{pmatrix}. \quad (4.65)$$

For $\lambda \neq 0$, the inverse relations are

$$E_+^{(6)} = 4\lambda (F_{tx}^{(6)})_{21}, \quad (4.66)$$

$$E_3^{(6)} = 4\lambda \left[(F_{tx}^{(6)})_{11} - 2\alpha_6 \lambda^2 (F_{tx}^{(6)})_{21} \right], \quad (4.67)$$

$$E_-^{(6)} = 4\lambda \left[(F_{tx}^{(6)})_{12} - 4\alpha_6 \lambda^2 (F_{tx}^{(6)})_{11} + 12\alpha_6^2 \lambda^4 (F_{tx}^{(6)})_{21} \right]. \quad (4.68)$$

Thus $F_{tx}^{(6)} = 0$ is equivalent to all three component equations (4.58)–(4.60) for every $\lambda \neq 0$ and every α_6 . This equivalence is a local identity on the unit-spin constraint surface.

5 Comparison of Class 5 and Class 6

The two finite-lattice calculations allow us to identify the common form of the surviving operator-product correction and the coefficients that depend on the model. We first express both corrections through the difference between the lower symbol of an operator square and the square of its lower symbol. We then compare the deformation scalings and spatial expansions, and evaluate the common isotropic limit.

5.1 Operator-product corrections in the coherent-state limit

The lower-symbol correction in both models originates in the algebra of operators acting on a common physical site. Equation (2.34) gives the difference between the symbol of a Pauli product and the product of the separate symbols. The spin representation remains fixed at spin $\frac{1}{2}$ as $\epsilon \rightarrow 0$, so this difference remains finite. Smoothness controls the variation of coherent-state labels between neighbouring sites. The operator products on their common site still obey Eq. (2.34).

In the discrete zero-curvature equation, the two overlapping products have connected contributions starting at order ϵ^2 . Their leading terms cancel at coincident spins. The first spatial Taylor coefficients and the order- ϵ^3 operator products then contribute at the same order as $\partial_{t_{\text{lat}}} \hat{L}$ under $t = \epsilon^2 t_{\text{lat}}$. Sections 3.2 and 4.2 evaluate this surviving contribution explicitly.

For a common expression of the result, let $m \in \{5, 6\}$ label the model and define the auxiliary-space matrix

$$\Xi_m := \left[(L_m^{(1)})^2 \right]^\downarrow - U_m^2, \quad U_m = (L_m^{(1)})^\downarrow. \quad (5.1)$$

All lower symbols here are evaluated at the same spin $\mathbf{S}(x, t)$. This definition extends the notation Ξ_6 in Eq. (4.35) to Class 5. For the two normalized chains studied here, the finite-lattice calculations give

$$\mathcal{C}_m = 2i(\partial_x \Xi_m + [\Xi_m, U_m]), \quad (5.2)$$

$$\Delta V_m = 2i\Xi_m - \frac{i}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a}, \quad V_m = V_m^\downarrow + \Delta V_m. \quad (5.3)$$

Equation (5.2) summarizes Eqs. (3.38) and (4.40). Substituting Eq. (5.3) into it gives $\mathcal{C}_m = \partial_x \Delta V_m + [\Delta V_m, U_m]$; the field-independent scalar leaves this identity unchanged and fixes the representatives used in the two model sections.

5.2 Common structure and model-dependent terms

The different deformation scalings follow from the bond lower symbols. For Class 5, the deformation vanishes at coincident spins and its leading term is proportional to $\kappa_5 \epsilon$. For Class 6, it already contributes at order κ_6 . Matching each contribution to the order- ϵ^2 exchange energy fixes $\kappa_5 = \epsilon \alpha_5$ and $\kappa_6 = \epsilon^2 \alpha_6$. The resulting Hamiltonian densities and spatial-operator coefficients are compared in Table 1. In the table, $L_5^{(2)} := 0$ denotes the absent second-order coefficient in the exact expansion (3.13). The matrices M_5 and N_6 are those in Eqs. (3.46) and (4.54).

	Class 5	Class 6
Lattice coupling	$\epsilon \alpha_5$	$\epsilon^2 \alpha_6$
Second spatial coefficient	0	$\frac{\alpha_6}{2} K_6 + \alpha_6^2 \lambda^2 K_6^2$
Operator square $(L_m^{(1)})^2$	$\frac{\mathbf{1}_4}{4\lambda^2}$	$\frac{\mathbf{1}_4}{4\lambda^2} + 2L_6^{(2)}$
Deformation term in the Hamiltonian density	$\frac{\alpha_5}{2} \mathbf{S}^\top M_5 \partial_x \mathbf{S}$	$-\frac{\alpha_6}{2} \mathbf{S}^\top N_6 \mathbf{S}$

Table 1: Comparison in the fixed exchange and time normalizations of Sections 3.1 and 4.1. The Hamiltonian densities share the exchange term $-\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S}$. The Class 5 total derivative has been removed using periodicity, as in Eq. (3.49).

The antisymmetric matrix M_5 couples the spin to its spatial derivative in Class 5. The symmetric matrix N_6 gives a local anisotropy term in Class 6. At the operator level, the distinction relevant to the time component is

$$\Xi_5 = \frac{\mathbf{1}_{\mathcal{V}_a}}{4\lambda^2} - U_5^2, \quad \Xi_6 = \frac{\mathbf{1}_{\mathcal{V}_a}}{4\lambda^2} + 2(L_6^{(2)})^\downarrow - U_6^2. \quad (5.4)$$

Thus $[\Xi_5, U_5] = 0$, and the Class 5 source is a total spatial derivative. For Class 6, the second spatial coefficient contributes both a derivative and a commutator through Eq. (4.50). The spatially uniform example (4.51) isolates the commutator contribution: it remains nonzero even when all spatial derivatives vanish.

The undeformed limit also retains an operator-product correction. To display it, write σ_a^i for the Pauli matrices acting on the auxiliary space $\mathcal{V}_a$. Setting $\alpha_m = 0$ in either model gives

$$U_m|_{\alpha_m=0} = \frac{\mathbf{1}_{\mathcal{V}_a} + S^i \sigma_a^i}{4\lambda}, \quad (5.5)$$

$$\Delta V_m|_{\alpha_m=0} = -\frac{iS^i \sigma_a^i}{4\lambda^2}, \quad \mathcal{C}_m|_{\alpha_m=0} = -\frac{i(\partial_x S^i) \sigma_a^i}{4\lambda^2}. \quad (5.6)$$

These formulas follow from $(S^i \sigma_a^i)^2 = \mathbf{1}_{\mathcal{V}_a}$ on the unit-spin constraint. Adding the correction to the direct time symbol gives the common isotropic Landau–Lifshitz time matrix

$$V_m|_{\alpha_m=0} = -\frac{(\mathbf{S} \times \partial_x \mathbf{S})^i \sigma_a^i}{2\lambda} + \frac{iS^i \sigma_a^i}{4\lambda^2}. \quad (5.7)$$

The direct symbol has twice the coefficient of the second term in Eq. (5.7). The surviving correction is therefore already present in the spin- $\frac{1}{2}$ XXX limit.

6 Summary and discussion

We have constructed the continuum time Lax matrices for the Class 5 and Class 6 Landau–Lifshitz models by taking the quantum lattice time operators to the long-wavelength limit. The lower symbols of products on overlapping physical sites yield a finite contribution to the continuum Lax equation. Its absorption into a local time correction gives zero-curvature equations equivalent to the independently derived Hamiltonian equations of motion. This equivalence follows from off-shell identities with explicit inverses. For Class 5, the time matrix also agrees with an independent classical r -matrix construction up to the addition of a field-independent multiple of the identity matrix.

The comparison of the two models identifies the operator terms required by this limit. The Class 5 contribution is a total spatial derivative, while the second spatial coefficient in Class 6 produces an additional commutator. A spatially uniform configuration isolates that commutator. The correction remains present in the undeformed XXX limit, showing that the effect also arises from the operator algebra of the isotropic chain. These results specify how the local quantum products determine the continuum time component and provide a direct connection between the lattice and classical constructions discussed in Refs. [5, 4].

The resulting Lax identities hold locally on the complex unit-spin constraint in the fixed spin- $\frac{1}{2}$ scaling. Periodicity is used for the Hamiltonian variation and the monodromy comparison. The relation to quantum dynamics concerns the instantaneous derivatives obtained by evaluating exact lattice commutators on product coherent states. An approximation to quantum evolution over a finite time interval would also require estimates for the correlations generated between sites and for the accumulated error of the continuum expansion.

A direct extension is to apply the same construction to higher lattice Hamiltonians. This would permit a comparison with the boost-generated continuum charges of Ref. [4] and the eleven-vertex hierarchy of Ref. [8]. As the time operators act on more physical sites, their overlapping products can involve additional operator moments. Determining which of these moments survive would test the range of the common formula (5.2). Studying the dependence on the local spin representation would further clarify how the long-wavelength limit combines with the classical limit of the spin algebra.

A Machine-learning searches for the Lax pairs

We determine coefficients in finite Lax ansätze by minimizing an off-shell residual, following the optimization approach of Ref. [11]. The input data are local values of the spin and its derivatives, $(\mathbf{S}, \partial_x \mathbf{S}, \partial_x^2 \mathbf{S}, \partial_t \mathbf{S})$. These are sampled subject to the unit-spin constraint and its differentiated identities, with the time derivative independent of the equations of motion. The sampling distributions are specified in Section A.3.

For each model $m \in \{5, 6\}$, a run holds the deformation parameter α_m and spectral parameter λ fixed. The equations of motion and the tensor structures allowed in the Lax ansatz are also supplied. The trainable parameters are the coefficients of those structures: thirteen real coefficients for Class 5 and eight for Class 6, each shared by all samples in a run. They are adjusted so that the curvature agrees with the equation-of-motion residual multiplied by the coefficient map determined from the spatial ansatz. This yields a candidate Lax pair at the chosen (α_m, λ) . Training requires no target values of a Lax pair. The numerical runs use real α_m and $\lambda \in i\mathbb{R}_{>0}$, for which the coefficient relations below are real. Their analytic verification applies to complex $\lambda \neq 0$.

To write the ansätze, represent traceless matrices by three-component vectors. With the Pauli matrices of Section 2, define

$$\iota(\mathbf{v}) := -\frac{i}{2} v^i \sigma^i, \quad [\iota(\mathbf{v}), \iota(\mathbf{w})] = \iota(\mathbf{v} \times \mathbf{w}). \quad (\text{A.1})$$

Write the candidate matrices as $U_m^{\text{fit}} = \mu_m \mathbf{1}_2 + \iota(\mathbf{u}_m)$ and $V_m^{\text{fit}} = \nu_m \mathbf{1}_2 + \iota(\mathbf{v}_m)$. Here $\mathbf{u}_m, \mathbf{v}_m$ contain the trainable coefficients, while the scalar parts μ_m, ν_m are prescribed functions, specified below. The superscript distinguishes the fitted candidate from the analytic matrices in the main text. The vector part of its curvature is

$$\mathcal{F}^{(m)} := \partial_t \mathbf{u}_m - \partial_x \mathbf{v}_m + \mathbf{u}_m \times \mathbf{v}_m. \quad (\text{A.2})$$

The vectors $\mathbf{E}^{(5)}$ and $\mathbf{E}^{(6)}$ denote $\partial_t \mathbf{S}$ minus the right-hand side of Eqs. (3.52) and (4.57), respectively. They are computed from the input data at each sampled point. All spin contractions and cross products are complex bilinear. In the losses, $\|\mathbf{v}\|^2 = \sum_i |v^i|^2$ is the positive Euclidean norm squared, and angle brackets denote the arithmetic mean over the training samples.

A.1 Class 5

The constant matrix M_5 in Eq. (3.46) can be written as $M_5 \mathbf{v} = \mathbf{m}_5 \times \mathbf{v}$, where

$$\mathbf{m}_5 = (i, -1, 0)^\top, \quad \mathbf{m}_5^\top \mathbf{m}_5 = 0, \quad M_5^2 = \mathbf{m}_5 \mathbf{m}_5^\top. \quad (\text{A.3})$$

The ansatz uses the spatial differential operator $\mathcal{D}_x \mathbf{v} := \partial_x \mathbf{v} + (\alpha_5/2) M_5 \mathbf{v}$ and the constant auxiliary-basis rotation

$$O_5(\lambda) := \exp(\alpha_5 \lambda M_5) = \mathbf{1}_3 + \alpha_5 \lambda M_5 + \frac{\alpha_5^2 \lambda^2}{2} M_5^2, \quad O_5^\top O_5 = \mathbf{1}_3, \quad \det O_5 = 1. \quad (\text{A.4})$$

Both $\mathcal{D}_x$ and O_5 are fixed by M_5 , α_5 and λ throughout a run.

Define the scalar contractions $\chi := \mathbf{m}_5^\top \mathbf{S}$, $\xi := \mathbf{m}_5^\top \mathcal{D}_x \mathbf{S}$, and $\varrho := \mathbf{m}_5^\top (\mathbf{S} \times \mathcal{D}_x \mathbf{S})$. The

twelve vector structures $\mathbf{W}_j$ used for the time component are

$$\begin{aligned}
\mathbf{W}_1 &= \mathbf{S} \times \mathcal{D}_x \mathbf{S}, & \mathbf{W}_2 &= \varrho \mathbf{m}_5, \\
\mathbf{W}_3 &= \mathbf{S}, & \mathbf{W}_4 &= \chi \mathbf{m}_5, \\
\mathbf{W}_5 &= \mathcal{D}_x \mathbf{S}, & \mathbf{W}_6 &= \chi \mathcal{D}_x \mathbf{S}, \\
\mathbf{W}_7 &= \xi \mathbf{S}, & \mathbf{W}_8 &= \mathbf{m}_5 \times \mathcal{D}_x \mathbf{S}, \\
\mathbf{W}_9 &= \chi(\mathbf{m}_5 \times \mathbf{S}), & \mathbf{W}_{10} &= \varrho \mathbf{S}, \\
\mathbf{W}_{11} &= \mathbf{m}_5, & \mathbf{W}_{12} &= \chi^2 \mathbf{m}_5.
\end{aligned} \tag{A.5}$$

The trainable coefficients are the spatial coefficient β_5 and the twelve temporal coefficients $\theta_1, \dots, \theta_{12}$. The candidate vectors are

$$\mathbf{u}_5 = O_5 \left(\frac{i}{2\lambda} \mathbf{S} + \beta_5 \chi \mathbf{m}_5 \right) - \frac{\alpha_5}{2} \mathbf{m}_5, \tag{A.6}$$

$$\mathbf{v}_5 = -2O_5 \sum_{j=1}^{12} \theta_j \mathbf{W}_j. \tag{A.7}$$

The coefficient of $\partial_t \mathbf{S}$ in $\partial_t \mathbf{u}_5$ determines the matrix

$$\mathbf{Q}_5 := O_5 \left(\frac{i}{2\lambda} \mathbf{1}_3 + \beta_5 \mathbf{m}_5 \mathbf{m}_5^\top \right). \tag{A.8}$$

Thus $\mathbf{Q}_5$ depends on the same β_5 as the spatial ansatz. For a prescribed sparsity weight $\eta > 0$ and smoothing scale $\delta = 10^{-10}$, the loss minimized over these thirteen coefficients is

$$\begin{aligned}
\mathcal{L}_5 &= \frac{1}{12} \left\langle \left\| O_5^{-1} (\mathcal{F}^{(5)} - \mathbf{Q}_5 \mathbf{E}^{(5)}) \right\|^2 \right\rangle \\
&\quad + \eta \left(\sqrt{\beta_5^2 + \delta^2} + \sum_{j=1}^{12} \sqrt{\theta_j^2 + \delta^2} \right).
\end{aligned} \tag{A.9}$$

The first term measures the curvature mismatch with a fixed, invertible weighting O_5^{-1} of its components. The second term is a smooth ℓ^1 penalty applied uniformly to all thirteen coefficients. The weight η is held fixed during each optimization stage, with the values given in Section A.3.

The prescribed scalar parts are

$$\begin{aligned}
\mu_5 &= \frac{1}{4\lambda} + \frac{\alpha_5}{4} S^+, \\
\nu_5 &= \frac{i\alpha_5}{2} (S^+ \partial_x S^3 - S^3 \partial_x S^+) + \frac{i\alpha_5^2}{4} (S^+)^2.
\end{aligned} \tag{A.10}$$

They obey $\partial_t \mu_5 - \partial_x \nu_5 = (\alpha_5/4) E_+^{(5)}$, where $E_+^{(5)}$ is defined in Eq. (3.57). The scalar curvature therefore already has the required factorization.

The numerical coefficients approach the relations

$$\begin{aligned}
\beta_5 &= -\frac{i\alpha_5^2 \lambda}{4}, & \theta_1 &= \frac{i}{2\lambda}, & \theta_2 &= \beta_5, \\
\theta_3 &= \frac{1}{4\lambda^2}, & \theta_4 &= \frac{\alpha_5^2}{8}, & \theta_5 &= \dots = \theta_{12} = 0.
\end{aligned} \tag{A.11}$$

Substitution gives exactly the matrices in Eqs. (3.18) and (3.41). The identity $\det \mathbf{Q}_5 = (i/(2\lambda))^3$ and the inverse relations (3.62) establish the local equivalence to the equations of motion.

A.2 Class 6

For Class 6, the known nilpotency $N_6^3 = 0$ of Eq. (4.54) determines the finite matrix basis $\{\mathbf{1}_3, N_6, N_6^2\}$. The eight real coefficients $a_1, a_2, b_0, b_1, b_2, c_0, c_1, c_2$ are the trainable parameters. They enter three coefficient matrices,

$$\begin{aligned} \mathbf{A}_6 &= \frac{i}{2\lambda} \mathbf{1}_3 + a_1 N_6 + a_2 N_6^2, \\ \mathbf{B}_6 &= b_0 \mathbf{1}_3 + b_1 N_6 + b_2 N_6^2, \\ \mathbf{C}_6 &= c_0 \mathbf{1}_3 + c_1 N_6 + c_2 N_6^2. \end{aligned} \quad (\text{A.12})$$

The coefficient $i/(2\lambda)$ of $\mathbf{1}_3$ in $\mathbf{A}_6$ is fixed throughout a run. The candidate vectors and scalar parts are

$$\begin{aligned} \mathbf{u}_6 &= \mathbf{A}_6 \mathbf{S}, & \mu_6 &= \frac{1}{4\lambda}, \\ \mathbf{v}_6 &= -2[\mathbf{B}_6 \mathbf{S} + \mathbf{C}_6(\mathbf{S} \times \partial_x \mathbf{S})], & \nu_6 &= 0. \end{aligned} \quad (\text{A.13})$$

Since $\partial_t \mathbf{u}_6 = \mathbf{A}_6 \partial_t \mathbf{S}$, the spatial ansatz fixes the coefficient map multiplying $\mathbf{E}^{(6)}$ to be $\mathbf{A}_6$. The loss minimized over the eight coefficients is

$$\mathcal{L}_6 = \frac{1}{12} \left\langle \left\| \mathcal{F}^{(6)} - \mathbf{A}_6 \mathbf{E}^{(6)} \right\|^2 \right\rangle, \quad (\text{A.14})$$

evaluated on the independently sampled spin derivatives. The nilpotency of N_6 gives $\det \mathbf{A}_6 = (i/(2\lambda))^3$ for every choice of the fitted coefficients, so its coefficient map is invertible for $\lambda \neq 0$.

The recovered coefficient relations are

$$\begin{aligned} a_1 &= c_1 = -i\alpha_6 \lambda, & a_2 &= c_2 = -i\alpha_6^2 \lambda^3, \\ b_0 &= \frac{1}{4\lambda^2}, & b_1 &= \frac{\alpha_6}{2}, \\ b_2 &= \frac{3\alpha_6^2 \lambda^2}{2}, & c_0 &= \frac{i}{2\lambda}. \end{aligned} \quad (\text{A.15})$$

They give $\mathbf{C}_6 = \mathbf{A}_6$ and the finite matrix-square-root identity

$$\mathbf{A}_6^2 = -\frac{1}{4\lambda^2} \mathbf{1}_3 + \alpha_6 N_6. \quad (\text{A.16})$$

The corresponding $U_6^{\text{fit}}, V_6^{\text{fit}}$ coincide with Eqs. (4.19) and (4.45). Their subsequent analytic verification gives $\mathcal{F}^{(6)} = \mathbf{A}_6 \mathbf{E}^{(6)}$ off shell, with $\det \mathbf{A}_6 = (i/(2\lambda))^3$. Thus the final equivalence to the equations of motion follows from an invertible coefficient map, as displayed explicitly in Section 4.3.

A.3 Sampling and numerical checks

The following distributions generate the input data. All dimensionful quantities are expressed in a fixed length unit. In Class 5, draw a real unit vector $\mathbf{S}$ by normalizing a three-dimensional standard Gaussian vector. Let $\Pi_{\mathbf{S}} := \mathbf{1}_3 - \mathbf{S}\mathbf{S}^T$ be its tangent projection, and draw independent standard real Gaussian vectors $\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_t$. The local derivatives are

$$\begin{aligned} \partial_x \mathbf{S} &= \Pi_{\mathbf{S}} \mathbf{g}_1 - \frac{\alpha_5}{2} M_5 \mathbf{S}, \\ \partial_x^2 \mathbf{S} &= \Pi_{\mathbf{S}} \mathbf{g}_2 - (\Pi_{\mathbf{S}} \mathbf{g}_1)^T (\Pi_{\mathbf{S}} \mathbf{g}_1) \mathbf{S} \\ &\quad - \alpha_5 M_5 \Pi_{\mathbf{S}} \mathbf{g}_1 + \frac{\alpha_5^2}{4} M_5^2 \mathbf{S}, \\ \partial_t \mathbf{S} &= -2\Pi_{\mathbf{S}} \mathbf{g}_t. \end{aligned} \quad (\text{A.17})$$

These data satisfy the unit-spin constraint and its differentiated identities; the time derivative is sampled independently of the equations of motion. Each run uses 768 training points and 4096 independent validation points. The pairs (α_5, λ) are $(0.8, 5i/7)$, $(0.8, 5i/12)$, and $(1.4, 5i/12)$, with three initializations for each pair.

For Class 6, draw a complex Gaussian vector $\mathbf{z}$, whose real and imaginary components have independent unit variance, and set $\mathbf{S} = \mathbf{z}/\sqrt{\mathbf{z}^\top \mathbf{z}}$. Samples with $|\mathbf{z}^\top \mathbf{z}| \leq 0.25$ are rejected. Using independent complex Gaussian vectors $\mathbf{g}_1, \mathbf{g}_2$ with the same distribution, set

$$\begin{aligned}\partial_x \mathbf{S} &= 0.8 \Pi_S \mathbf{g}_1, \\ \partial_x^2 \mathbf{S} &= 0.8 \Pi_S \mathbf{g}_2 - (\partial_x \mathbf{S})^\top (\partial_x \mathbf{S}) \mathbf{S}.\end{aligned}\tag{A.18}$$

For the off-shell loss, an independent complex Gaussian vector $\mathbf{g}_t$ with the same distribution gives $\partial_t \mathbf{S} = -2 \Pi_S \mathbf{g}_t$. Each run uses 768 training points and 3072 independent validation points. The pairs (α_6, λ) are $(0.35, 10i/13)$, $(0.7, i/2)$, and $(1.1, 10i/29)$, with two initializations each.

Class 5 starts with independent zero-mean Gaussian coefficients of standard deviation 0.25. Class 6 starts from the undeformed values $b_0 = 1/(4\lambda^2)$, $c_0 = i/(2\lambda)$, and zero for the remaining coefficients, with independent zero-mean real Gaussian perturbations of standard deviation 0.08. Both searches use Adam followed by the limited-memory Broyden–Fletcher–Goldfarb–Shanno method (L-BFGS). For Class 5, Adam has 600 steps and learning rate 0.03; L-BFGS has at most 50 iterations and learning rate 0.8. The sparsity weight in Eq. (A.9) is $\eta = 10^{-9}$ during Adam and 5×10^{-11} during L-BFGS. For Class 6, Adam has 1300 steps and learning rate 0.02; L-BFGS has at most 120 iterations and learning rate 0.7. At every step, the loss is evaluated on the full, fixed training set, and its gradients update the ansatz coefficients. Both searches use double precision. After optimization, the same residuals are evaluated on independently generated validation samples; the sparsity penalty is excluded from the Class 5 validation residual.

Across the runs specified above, the largest absolute coefficient error was 3.61×10^{-7} in Class 5 and 1.12×10^{-11} in Class 6. The largest absolute coefficient of $\mathbf{W}_5, \dots, \mathbf{W}_{12}$ was 8.10×10^{-8} . The maximum validation mean-square residual in the first term of Eq. (A.9) was 4.94×10^{-14} . For Class 6, the maximum validation root-mean-square residual, $\sqrt{\mathcal{L}_6}$, was 2.92×10^{-11} . The analytic coefficient relations were used only for evaluation after optimization. Their exact substitution into the Lax matrices and the off-shell identities in Sections 3.3 and 4.3 establish the physical results. The evaluation of overlapping quantum products in the main text supplies the further connection to the finite lattice time operators.

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